HPSC & HSSC SCREENING EXAM COMPREHENSIVE DOSSIER

# HARYANA GENERAL KNOWLEDGE

Master Preparation Keynote for Food Safety Officer (FSO), HCS & Screening Tests

### I. HISTORY & ARCHAEOLOGY

Vedic Roots, Rakhigarhi, 3 Battles of Panipat, 1857 Heroes & Chhotu Ram

### II. GEOGRAPHY & HYDROLOGY

Shivalik to Aravalli, Yamuna-Ghaggar, Canals, Soils, Forests & Sanctuaries

### III. AGRICULTURE & FOOD ECOSYSTEM

FSO Special: Murrah Buffalo, NDRI, NIFTEM, Mega Food Parks & Crop Belts

### IV. POLITY, ECONOMY & DISTRICTS

90 Vidhan Sabha, Census 2011, 22 Districts Dossier & Top 30 Exam Traps

*Based on Comprehensive Cross-Examination Analysis of 10+ Years HPSC / HSSC Screening Question Papers*

SCREENING EXAM BLUEPRINT

# Weightage & Question Distribution in Haryana Screening Tests

*Key Exam Takeaway: Haryana GK carries 20-25% decisive weightage in HPSC FSO and ADO screening tests.*

| Exam Category | Conducting Body | Total Questions | Haryana GK Weightage | Core High-Frequency Focus Areas |
|---|---|---|---|---|
| **Food Safety Officer (FSO)** | HPSC | 100 MCQs | 20-25 Questions | Agriculture, Murrah Buffalo, NDRI/NIFTEM, Food Laws, Soil & Geography |
| **HCS (Executive Branch)** | HPSC | 100 MCQs (Paper 1) | 15-20 Questions | Ancient History, Harappan Sites, 1857 Revolt, Vidhan Sabha Polity, Schemes |
| **Agriculture Development Officer** | HPSC | 100 MCQs | 20 Questions | Soils, Canal Irrigation (WYC/Bhakra), Crop Seasons, Agro-Climatic Zones |
| **Assistant Professor (College)** | HPSC | 100 MCQs | 15-20 Questions | History, Folk Literature (Saang), Census 2011, Demographics, Personalities |
| **HSSC CET (Group C / D)** | HSSC | 100 MCQs | 25 Questions (Fixed 25%) | District Profiles, Folk Dances, Rivers, Sports Stars, Wildlife Sanctuaries |

### FSO SCREENING STRATEGIC MANDATE

In the Food Safety Officer (FSO) screening test, subject knowledge (Food Tech, Microbiology, Chemistry, FSS Act 2006) accounts for ~80% of technical marks, while Haryana GK forms the mandatory non-technical section (20%). Candidate qualification cutoff hinges on this section because technical scores are closely grouped. High-yield topics include: Dairy Research Infrastructure (NDRI, CIRB, LUVAS), Basmati Rice & Mustard Zones, Mega Food Parks, Soil Salinity Reclamation (CSSRI), and Government Agro-Nutrition Initiatives.

Analysis of HPSC FSO 2023, HPSC ADO 2022-23, HPSC HCS Pre 2021-2024, and HSSC CET Group C/D Exam Patterns.

# The Political Journey to Statehood: 1858 to 1966

*Key Exam Takeaway: Haryana was tagged to Punjab in 1858 as British collective punishment for active participation in the 1857 revolt.*

## PHASE I: PRE-INDEPENDENCE DEMANDS

• **Colonial Annexation (1858):** After the 1857 First War of Independence, the British detached Haryana from North-Western Provinces and merged it with Punjab to penalize the rebellious peasantry.

• **First Formal Demand (1926):** All India Muslim League Delhi session, Pirzada Muhammad Hussain demanded separation of Delhi and Haryana territory from Punjab.

• **Deshbandhu Gupta's Advocacy (1932):** Formulated a clear plea for 'Greater Delhi' integrating Ambala Division, Meerut, and Agra into a culturally unified linguistic zone.

• **Dinbandhu Sir Chhotu Ram:** Stood for peasant rights through the Unionist Party (1923), safeguarding rural agrarian interests under the Punjab Legislative Council.

## PHASE II: POST-INDEPENDENCE FORMULAS

• **Sachar Formula (1 Oct 1949):** Proposed by CM Bhim Sen Sachar. Divided Punjab into 'Punjabi Zone' (Gurmukhi script) and 'Hindi Zone' (Devanagari script). Rejected due to language friction.

• **Fazal Ali Commission (SRC 1953-55):** States Reorganisation Commission rejected separate Haryana, citing administrative synergy, causing massive discontent among Hindi speakers.

• **Regional Formula (1956):** Established Hindi and Punjabi Regional Committees in Punjab Assembly. Failed to resolve deep developmental and cultural disparities.

• **Sant Fateh Singh's Agitation (1965):** Threat of self-immolation for Punjabi Suba forced Central Government to form Parliamentary Committee headed by Lok Sabha Speaker Hukam Singh.

# Shah Commission & 18th Amendment: Birth of 17th State

*Key Exam Takeaway: Haryana was created under the Punjab Reorganisation Act 1966 via the 18th Constitutional Amendment.*

## SHAH COMMISSION KEY PARAMETERS

• **Constitution Date:** 23 April 1966 by the Government of India under the chairmanship of Justice J.C. Shah (Supreme Court Judge).

• **Commission Members:** Justice J.C. Shah (Chairman), S. Dutt (Member), and M.M. Philip (Member).

• **Mandate:** Demarcate the Hindi-speaking areas of the composite State of Punjab to carve out a new State.

• **Report Submission:** 31 May 1966. Recommended creation of Haryana with Ambala, Gurgaon, Hissar, Rohtak, Mahendragarh, Jind, and Narwana tehsils.

• **Kharar & Chandigarh Controversy:** Commission recommended Kharar tehsil (including Chandigarh) to Haryana by 2:1 majority (S. Dutt dissented). Center later made Chandigarh a Joint Capital Union Territory.

## CONSTITUTIONAL & LEGISLATIVE FRAMEWORK

• **18th Constitutional Amendment Act 1966:** Clarified parliamentary power under Article 3 to form a new State or Union Territory by uniting parts of existing states.

• **Punjab Reorganisation Act 1966:** Passed by Parliament in September 1966; received Presidential assent on 18 September 1966.

• **Official Formation:** 1 November 1966 as the 17th State of the Republic of India.

• **Area Ceded:** Approximately 35.18% of the composite Punjab area (44,212 sq km) was transferred to Haryana.

• **Shared Institutions:** Common High Court (Punjab & Haryana High Court at Chandigarh), Joint Capital (Chandigarh), and shared Bhakra Beas Management Board (BBMB).

ADMINISTRATIVE EVOLUTION

# District Genesis: From Original 7 ('GHAR JK') to 22 Districts

*Key Exam Takeaway: Haryana began on Nov 1, 1966 with 7 districts; Charkhi Dadri was carved out as the 22nd district in Dec 2016.*

| Chronological Phase | Districts Created | Carved Out From | Formation Date | Exam Significance |
|---|---|---|---|---|
| **Original 7 (At Birth)** | Gurgaon, Hissar, Ambala, Rohtak, Jind, Karnal, Mahendragarh | Punjab Reorganisation Act | 1 Nov 1966 | Acronym 'GHAR JK'. Largest at birth: Hissar; Smallest: Jind. |
| **Phase 2 (1972-73)** | Bhiwani & Sonipat (1972)<br>Kurukshetra (1973) | Hissar/Rohtak<br>Karnal | 22 Dec 1972<br>23 Jan 1973 | Bhiwani created by Bansi Lal; Kurukshetra named after King Kuru. |
| **Phase 3 (1975-79)** | Sirsa (1975)<br>Faridabad (1979) | Hissar<br>Gurgaon | 1 Sep 1975<br>2 Aug 1979 | Sirsa became largest district after Bhiwani split. Faridabad industrial hub. |
| **Phase 4 (1989 Quad)** | Kaithal, Panipat, Rewari, Yamunanagar | Karnal, Rohtak, Gurgaon, Ambala | 1 Nov 1989 | Created simultaneously on Haryana Day. Panipat merged back & re-created 1992! |
| **Phase 5 (1995-97)** | Panchkula (1995)<br>Fatehabad & Jhajjar (1997) | Ambala<br>Hissar & Rohtak | 15 Aug 1995<br>15 Jul 1997 | Panchkula planned satellite; Fatehabad historical city of Firoz Shah. |
| **Phase 6 (2005-16)** | Nuh/Mewat (2005), Palwal (2008), Charkhi Dadri (2016) | Gurgaon/Faridabad<br>Bhiwani | 4 Apr 2005<br>15 Aug 2008<br>1 Dec 2016 | Nuh created as Satyamevpuram. Charkhi Dadri is the 22nd & newest district. |

STATE IDENTITY

# Official State Symbols & Emblems of Haryana

*Key Exam Takeaway: Black Francolin (Kala Teetar) is State Bird; Peepal is State Tree; Lotus is State Flower.*

## STATE BIRD: BLACK FRANCOLIN

- **Scientific Name:** Francolinus francolinus.
- **Local Name:** Kala Teetar (काला तीतर).
- **Habitat:** Shrublands, cultivated fields of Sirsa, Hisar, Bhiwani, and Kurukshetra.
- **Ecological Role:** Natural pest predator in wheat and mustard fields.
- **HPSC Trap:** Do not confuse with Great Indian Bustard (Rajasthan) or Monal (Uttarakhand).

## STATE ANIMAL: BLACKBUCK

- **Scientific Name:** Antilope cervicapra.
- **Local Name:** Krishna Mriga / Kala Hiran (काला हिरण).
- **Primary Range:** Abubshahar Wildlife Sanctuary (Sirsa), Bishnoi settlements in Fatehabad and Hisar.
- **Cultural Protector:** Bishnoi community upholds absolute sacred protection.
- **IUCN Status:** Least Concern (Schedule I of Wildlife Protection Act 1972).

## STATE TREE: PEEPAL

- **Scientific Name:** Ficus religiosa.
- **Local Name:** Peepal (पीपल) / Bodhi Vriksha.
- **Sacred Significance:** Mentioned in Rigveda and Bhagavad Gita (Lord Krishna calls himself Ashvattha among trees).
- **Medicinal & Oxygen Value:** Continuous oxygen producer; rich in flavonoids and tannins used in traditional medicine.
- **HPSC Trap:** State Tree is Peepal, NOT Khejri (which is Rajasthan's State Tree).

## FLOWER, SEAL & LANGUAGE

- **State Flower:** Lotus (Nelumbo nucifera) / Kamal. Sacred aquatic herb prevalent in village ponds and lakes.
- **Official State Seal:** Circular seal featuring Lion Capital of Ashoka with a rising sun emerging behind lotus blooms, flanked by ears of wheat.
- **Official Language:** Hindi (Devanagari script). Punjabi was declared 2nd official language in 2010 (under Bhupinder Singh Hooda; earlier Bansi Lal made Tamil 2nd language in 1969 to counter Punjabi).
- **State Sport:** Wrestling (Kushti) - Traditional soil akhadas producing 60%+ of national Olympic wrestling medals.

Haryana Government Notification No. 12/4/85-Edu; State Wildlife Wing.

GEOGRAPHICAL METRICS

# Geographical Coordinates, Territorial Extent & Border States

*Key Exam Takeaway: Haryana covers 44,212 sq km (1.34% of India) and is landlocked by 4 States & 2 UTs.*

## COORDINATES & PHYSICAL EXTENT

- **Latitudinal Extent:** 27°39' N to 30°55' N latitude (Span: ~3°16').
- **Longitudinal Extent:** 74°27' E to 77°36' E longitude (Span: ~3°09').
- **Total Geographical Area:** 44,212 sq km (17,070 sq miles), accounting for 1.34% of India's total land area.
- **National Area Rank:** 20th among Indian States (21st including UT of Ladakh/J&K reorganisation).
- **Elevation Profile:** Ranging from 200 m (656 ft) in southwestern depressions to 1,514 m in the northern Morni Hills.

## BOUNDARIES & BORDER DYNAMICS

- **Bordering States (4 States):** Punjab (North-West), Himachal Pradesh (North), Uttar Pradesh (East), and Rajasthan (South & South-West).
- **Bordering Union Territories (2 UTs):** New Delhi (surrounded on three sides: North, West, South) and Chandigarh (Joint Capital in North).
- **Longest Inter-State Border:** Rajasthan shares the longest boundary (~1,262 km) with 7 Haryana districts (Sirsa, Fatehabad, Hisar, Bhiwani, Mahendragarh, Rewari, Nuh).
- **Shortest Inter-State Border:** Himachal Pradesh shares ~11 km border with Panchkula, Ambala, and Yamunanagar.
- **Landlocked Nature:** Completely enclosed with zero international border and no coastal outlet.
- **Districts with No Inter-State Boundary:** Rohtak and Charkhi Dadri (touch only internal Haryana districts).

PHYSIOGRAPHIC ZONING

# Physiography of Haryana: Dr. Jasbir Singh's 8-Fold Classification

*Key Exam Takeaway: In 'Agricultural Geography of Haryana', Dr. Jasbir Singh classified the terrain into 8 distinct relief units.*

| Physiographic Unit | Geographical Coverage | Topographical Nature | Key Soil / Ecological Feature |
|---|---|---|---|
| **1. Shivalik Hills** | Panchkula, Northern Ambala & Yamunanagar | Tertiary folded hills; altitude 400m - 1514m | Subject to severe gully erosion; Pinjore gravel zone |
| **2. Piedmont Plain (Ghar/Kandi)** | Foothills of Shivaliks (Panchkula, Ambala) | Piedmont alluvial fans formed by seasonal torrents | Coarse sand, boulders; dissected by 'Chos' (torrents) |
| **3. Alluvial Plain (Bangar)** | Central Haryana (Kaithal, Karnal, Jind, Rohtak) | Flat, older alluvium plain; elevation 220m - 250m | Highly fertile loam; heart of Green Revolution |
| **4. Flood Plains (Khadar/Naili/Bet)** | Yamuna, Ghaggar, and Markanda flood belts | Low-lying newer alluvium formed by seasonal overflows | Khadar (Yamuna), Naili (Ghaggar), Bet (Markanda) |
| **5. Sandy Plains & Sand Dunes** | Sirsa, Fatehabad, Hisar, Bhiwani, Mahendragarh | Aeolian undulating plains with shifting sand dunes (Tibas) | Semi-arid; high wind erosion; calcareous dry soils |
| **6. Aravalli Outliers** | Gurugram, Mewat (Nuh), Rewari, Mahendragarh | Denuded Precambrian residual metamorphic ridges | Quartzite, slate, granite outcrops; rocky thin soils |
| **7. Drifting Sand Zone (Taal)** | Southwestern desert transition belt (Bhiwani/Sirsa) | Shallow depressions between sand dunes | Saline water accumulation during episodic rain |
| **8. Marshy / Tarai Tract** | Southern Sirsa (Sanisa depression / seasonal marshes) | Low-lying water accumulating basins | Heavy clayey soils; prone to temporary waterlogging |

# The Shivalik Range, Morni Hills & Karoh Peak Profile

*Key Exam Takeaway: Karoh Peak (1,514m) in Morni Hills, Panchkula is the highest elevation point of Haryana.*

## SHIVALIK RELIEF & MORNI HILLS

• **Geological Origin:** Outer-most Himalayan foothills formed during late Cenozoic era by fluvial deposits folded during Himalayan orogeny.

• **Hypsometric Peak:** Karoh Peak (करोह चोटी) situated in Morni Hills, Panchkula district at an elevation of 1,514 metres (4,967 ft) - widely asked in HPSC/HSSC.

• **Morni Hills Ecosystem:** The only hill station of Haryana, home to Tikkar Taal (twin lakes: Chhota Tikkar and Bada Tikkar separated by hillock).

• **Drainage Origin:** Cradle of seasonal foothill streams: Tangri, Ghaggar tributaries, and Markanda torrents.

• **Vegetation:** Sub-tropical chir pine (Pinus roxburghii), mixed deciduous forests, and rich wild medicinal flora.

## GHAR, KANDI & THE CHERRAPUNJI OF HARYANA

• **Piedmont Plain (Ghar & Kandi):** Narrow alluvial strip (10-15 km wide) lying immediately south of Shivaliks across Panchkula, Ambala, and Yamunanagar.

• **Local Terminology Trap:** Known locally as 'Ghar' (घर) in Kalka/Raipur Rani and 'Kandi' (कांडी) in hilly margins.

• **Torrents ('Chos'):** Cut deep ravines and seasonal dry beds called Chos that cause suddenflash floods during monsoon.

• **Chhachhrauli Phenomenon:** Located in Yamunanagar district; receives highest average annual rainfall in the state (>1100 mm), famously dubbed 'The Cherrapunji of Haryana'.

• **Geological Assets:** Boulder gravel, sandstone beds, and lime mortar kankar deposits extensively quarried for regional construction.

# Aravalli Outliers, Residual Metamorphic Ridges & Dhosi Hill

*Key Exam Takeaway: Dhosi Hill (652m) in Mahendragarh is the highest peak of Aravallis in Haryana and the hermitage of Rishi Chyavana.*

## DHOSI HILL & ARAVALLI RIDGES

• **Precambrian Metamorphic Antiquity:** One of the oldest fold mountain chains in the world (Delhi Supergroup: Alwar and Ajabgarh series).

• **Geographical Spread:** Discontinuous relict hills entering through Rajasthan into Mahendragarh, Rewari, Gurugram, Nuh (Mewat), and Faridabad.

• **Highest Elevation Peak:** Dhosi Hill (धोसी की पहाड़ी) at 652 metres (2,139 ft) situated at the border of Mahendragarh (near Narnaul) and Jhunjhunu (Rajasthan).

• **Extinct Volcanic Crater:** An isolated volcanic plug of late Proterozoic era with a crater-like depression at the summit.

• **Ecological Shield:** Acts as a critical green natural barrier preventing eastward expansion of the Thar Desert into Indo-Gangetic plains.

## HISTORICAL, MEDICINAL & ECOLOGICAL ROLE

• **Vedic Hermitage of Chyavana Rishi:** Ancient seat where sage Chyavana (son of sage Bhrigu) formulated the legendary Ayurvedic preparation 'Chyawanprash' using wild Amla and herbs.

• **Surajkund Sun Temple & Amphitheatre:** Built by Tomar King Anangpal I in 10th century CE in the Aravalli foothills of Faridabad.

• **Kotla & Khalilpur Depressions:** Natural drainage basins at the foot of Mewat Aravalli hills collecting seasonal runoff.

• **Mining & Conservation Ban:** Supreme Court of India banned commercial mining in Aravalli ridges across Faridabad and Gurugram to restore groundwater aquifers.

• **Aravalli Green Wall Project:** Ambitious national 1,400 km long and 5 km wide afforestation green belt spanning Haryana, Rajasthan, Gujarat, and Delhi.

Archaeological Survey of India (ASI); Gazetteers of Gurgaon & Mahendragarh.

# Soil Classification, Texture Profile & Agro-Ecological Zones

*Key Exam Takeaway: Soils vary from light sandy loams in the south-west to fertile alluvial loams in central plains and rocky loams in Shivaliks.*

## SOIL TEXTURAL CLASSES

• **Very Light Soils (Sandy):** Found in Sirsa, Fatehabad, Bhiwani, and Mahendragarh. High permeability, deficient in nitrogen and organic matter, prone to wind erosion.

• **Light Soils (Sandy Loam):** Spans dry parts of Hisar, Rewari, and Jhajjar. Highly responsive to tubewell/canal irrigation and phosphatic fertilizers.

• **Medium Soils (Loam & Silt Loam):** Dominant in Kaithal, Jind, Sonipat, Panipat, and Rohtak. The backbone of wheat, basmati paddy, and sugarcane cropping.

• **Moderately Heavy & Heavy Soils (Clayey):** Trapped along Ghaggar basin (Naili tract) and parts of Yamunanagar and Karnal; rich in nutrients but slow drainage.

## AGRO-ECOLOGICAL DISTRIBUTION

• **Agro-Climatic Zone I (Eastern / Wet Zone):** Panchkula, Ambala, Yamunanagar, Kurukshetra, Karnal, Kaithal, Sonipat, Panipat. Sub-humid to semi-arid, high irrigation, paddy-wheat rotation.

• **Agro-Climatic Zone II (Western / Dry Zone):** Sirsa, Fatehabad, Hisar, Bhiwani, Charkhi Dadri, Mahendragarh, Rewari, Jhajjar. Arid to semi-arid, mustard-pearl millet-cotton cultivation.

• **Organic Carbon Status:** Generally low to medium (<0.5%) due to intensive double-cropping and crop residue burning.

• **Micronutrient Deficiencies:** Zinc (Zn) and Iron (Fe) widespread across central plains; Sulphur (S) deficit pronounced in oilseed mustard belts.

# Soil Salinity, Alkalinity (Kallar / Rehi) & CSSRI Reclamation

*Key Exam Takeaway: CSSRI Karnal pioneered gypsum application and sub-surface drainage (SSD) to reclaim sodic soils.*

| Diagnostic Property | Saline Soils (Solonchak) | Alkali / Sodic Soils (Kallar/Rehi) | CSSRI Reclamation Technology |
|---|---|---|---|
| **pH Range** | < 8.5 (Neutral to mildly alkaline) | > 8.5 (Highly alkaline, up to 10.5) | Gypsum ($CaSO_4.2H_2O$) requirement based on Soil Testing |
| **Electrical Conductivity (EC)** | > 4.0 dS/m (High soluble salts) | < 4.0 dS/m (Low neutral soluble salts) | Leaching with good quality canal water through deep furrows |
| **Exchangeable Sodium (ESP)** | < 15% (Low sodium saturation) | > 15% (Excessive exchangeable $Na^+$) | Sub-Surface Drainage (SSD) with corrugated perforated PVC pipes |
| **Physical Soil Structure** | Flocculated, good aeration, permeable | Dispersed, crusting, impermeable hard pan | Growing salt-tolerant crops: CSR-30 Basmati, KRL-210 Wheat |
| **Major Affected Districts** | Rohtak, Jhajjar, Hisar, Jind, Sonipat | Karnal, Kurukshetra, Kaithal, Panipat | Bio-drainage using Eucalyptus plantation along canal seepage |

ICAR-Central Soil Salinity Research Institute (CSSRI), Karnal (Est. 1969); Haryana Reclamation Corporation.

HYDROLOGICAL DIVIDE

# River Systems of Haryana: Northern & Southern Drainage Networks

*Key Exam Takeaway: Haryana forms the great hydrological water divide between the Indus Basin and the Ganga Basin.*

## NORTHERN DRAINAGE SYSTEM (SHIVALIK ORIGIN)

• **Yamuna River:** The only perennial river; marks the entire eastern border with Uttar Pradesh; drains into the Bay of Bengal (Ganga System).

• **Ghaggar River:** Main seasonal river of northern Haryana; originates in Dagshai (HP); passes into Rajasthan desert (Indus / Inland System).

• **Markanda River:** Ancient Aruna; originates in Paonta/Dharti Dhar hills (HP); famous for seasonal fury and heavy siltation.

• **Tangri (Dangri) River:** Originates in Morni Hills; joins Markanda near Mullana (Ambala).

• **Saraswati River:** Ancient sacred river; originating near Adi Badri (Yamunanagar); revived by the Haryana Saraswati Heritage Development Board.

## SOUTHERN DRAINAGE SYSTEM (ARAVALLI ORIGIN)

• **Sahibi River:** Chief river of southern Haryana; originates in Kotputli/Bairath hills (Rajasthan); flows through Rewari and Gurugram into Najafgarh drain.

• **Krishnavati (Kasaunti) River:** Originates in Neem Ka Thana (Rajasthan); enters Mahendragarh; historically joins Sahibi.

• **Dohan River:** Originates near Mandoli (Rajasthan); mythological origin from the eye of Rishi Bhrigu's wife Divyapoloma; seasonal dry stream in Mahendragarh.

• **Indori River:** Originates in Nuh hills near old fort of Indore; divides into two branches near Rewari and meets Sahibi.

• **Inland Drainage:** Southern streams disappear into the sandy tracts of Mahendragarh and Rewari without reaching any sea.

PERENNIAL WATERWAY

# The Yamuna River: Entry, Course, Barrages & Basin Dynamics

*Key Exam Takeaway: Yamuna enters Haryana at Kalesar (Yamunanagar) and exits at Hasanpur (Palwal), spanning ~320 km.*

## RIVER COURSE & DEMARCATION

• **Glacial Source:** Originates from Yamunotri Glacier at Bandarpunch peak in Uttarkashi district (Uttarakhand) at ~6,387 m elevation.

• **Point of Entry:** Enters Haryana through Kalesar National Park forest in Yamunanagar district.

• **Natural Boundary Line:** Flows continuously southward defining the ~320 km interstate border between Haryana (Yamunanagar, Karnal, Panipat, Sonipat, Faridabad, Palwal) and Uttar Pradesh.

• **Point of Exit:** Leaves Haryana near Hasanpur village in Palwal district to flow into Mathura/Agra (UP).

• **Floodplain (Khadar):** Creates fertile flood-deposited newer alluvium belt across eastern Yamunanagar and Karnal.

## BARRAGES, DIVERSIONS & PROJECTS

• **Tajewala Barrage:** Historical stone diversion structure constructed by the British in 1873 to divert water into Western and Eastern Yamuna Canals.

• **Hathnikund Barrage:** Modern concrete barrage constructed in 1999 (operational 2002) 3 km upstream of Tajewala to handle super-floods (>8 lakh cusecs).

• **Okhla Barrage:** Located in Delhi/Faridabad border; feeds Agra Canal and Gurgaon Canal providing irrigation to southern districts.

• **Kishau, Renuka & Lakhwar Dams:** Upstream storage projects on tributaries (Tons, Giri, Yamuna) designed to ensure lean-season flow to Haryana.

• **FSO Ecological Note:** High industrial effluent contamination from Sonipat-Panipat-Gurugram drains (Dhanauri, Najafgarh) impacting downstream irrigation water.

# Ghaggar, Markanda (Aruna) & The Saraswati Heritage Project

*Key Exam Takeaway: Ghaggar creates the 'Naili' flood plain; Markanda merges into Saraswati at Sanisa Lake near Pehowa.*

| River Parameter | Ghaggar River | Markanda River (Aruna) | Saraswati River Project |
|---|---|---|---|
| **Origin Point** | Dagshai hills (Himachal Pradesh) at ~1,927m | Dharti Dhar / Paonta hills (HP) | Adi Badri foothills (Yamunanagar district) |
| **Entry into Haryana** | Near Kalka (Panchkula district) | Near Ambala district border | Adi Badri / Kathgarh cultural corridor |
| **Districts Traversed** | Panchkula, Ambala, Kaithal, Fatehabad, Sirsa | Ambala and Kurukshetra | Yamunanagar, Kurukshetra, Kaithal |
| **Major Tributaries** | Tangri, Markanda, Kaushalya (dam at Pinjore) | Runn, Begna, Nakti, and Tangri river | Chautang (ancient Drishadvati) historically |
| **Key Reservoirs / Sinks** | Ottu Barrage / Weir (Sirsa); dries in Thar Desert | Sanisa Lake (near Pehowa / Kurukshetra) | Palaeochannel excavations by HSHDB & ONGC |
| **Exam Terminology** | Flood plain called 'Naili'; non-perennial | Known as 'Aruna' in Mahabharata & Puranas | Mughalwali (Yamunanagar) site of active water revival |

Haryana Saraswati Heritage Development Board (HSHDB); State Remote Sensing Application Centre (HARSAC).

# Southern River Systems: Sahibi, Krishnavati, Dohan & Indori

*Key Exam Takeaway: Sahibi is the largest southern river, entering near Rewari and discharging into Najafgarh Jheel.*

## SAHIBI RIVER & MASANI BARRAGE

- **Chief Southern Artery:** Originates in Bairath / Sewar hills in Jaipur district, Rajasthan.
- **Entry into Haryana:** Enters Rewari district near Kotkasim border, flows through Pataudi (Gurugram) and Jhajjar.
- **Discharge Sink:** Empties into Najafgarh drain (historical Najafgarh lake) in Southwest Delhi.
- **Masani Barrage:** Built across Sahibi river in 1989 near Masani village (Rewari) on NH-48 for flood moderation and groundwater recharge.
- **Sub-Basin Significance:** Recharges vital deep aquifers in the water-stressed arid Rewari-Gurugram industrial belt.

## DOHAN, KRISHNAVATI & INDORI RIVERS

- **Dohan River:** Mythologically linked to Rishi Bhrigu's hermitage at Dhosi Hill; originates near Mandoli village; flows northward into Mahendragarh before getting lost in shifting sands.
- **Krishnavati (Kasaunti) River:** Originates from Dariba hills in Neem Ka Thana (Rajasthan); enters south of Narnaul; flows past Mahendragarh to disappear in Jhajjar depressions.
- **Indori River:** Takes birth near ancient ruined fort of Indore in Nuh (Mewat) hills; splits into two courses near Malab, one joining Sahibi near Rewari.
- **Ephemeral Nature:** All four streams are completely non-perennial and flow only during extreme cloudbursts in monsoon season.
- **Groundwater Depletion Alert:** Entire basin categorized as 'Over-exploited' / 'Dark Zone' by CGWB due to extensive tubewell extraction.

# Canal Irrigation Networks: Western Yamuna Canal & Bhakra System

*Key Exam Takeaway: WYC is Haryana's oldest canal (1355 CE); Tohana in Fatehabad is renowned as the 'City of Canals'.*

## WESTERN YAMUNA CANAL (WYC)

• **Historical Antiquity:** Originally excavated by Sultan Firoz Shah Tughlaq in 1355 CE to supply water to hunting lodges in Hisar.

• **Mughal Restoration:** Reconditioned by Akbar (1568) and remodeled as Nahr-i-Bihisht by Shah Jahan under Ali Mardan Khan to bring water to Delhi.

• **British Modernization:** Fully rebuilt in 1879 originating from Tajewala Barrage (now Hathnikund).

• **Branches & Command:** Main Branch divides into Delhi Branch, Hansi Branch, Bhalaut Branch, Barwala Branch, and Sunder Branch.

• **Command Districts:** Irrigates Yamunanagar, Karnal, Kaithal, Rohtak, Sonipat, Panipat, and Jind.

## BHAKRA SYSTEM & LIFT IRRIGATION SCHEMES

• **Bhakra Nangal Canal System:** Originates from Nangal Dam on Sutlej river; enters Haryana at Tohana (Fatehabad district) - known as 'Canal City'.

• **Main Branches:** Narwana Branch and Barwala Link Canal connecting Bhakra waters to Western Yamuna Canal system.

• **Jawaharlal Nehru (JLN) Lift Canal:** Multi-stage lift irrigation system engineered under CM Bansi Lal to pump water uphill to arid sandy terrains of Mahendragarh and Rewari.

• **Loharu (Indira Gandhi) & Siwani Lift Canals:** Lift canal systems designed to irrigate sand dune tracts of Bhiwani and Charkhi Dadri.

• **SYL Canal Dispute:** 214 km Sutlej-Yamuna Link canal (122 km in Punjab, 92 km in Haryana) constructed by Haryana but blocked by Punjab.

AQUATIC GEOGRAPHY

# Prominent Lakes, Wetland Reservoirs & Sacred Sarovars

*Key Exam Takeaway: Damdama is the largest natural lake; Brahma Sarovar is the sacred epicenter of Solar Eclipse pilgrimages.*

| Water Body / Sarovar | District Location | Origin / Structural Type | Key Exam Fact / Screening Point |
|---|---|---|---|
| **Damdama Lake** | Sohna (Gurugram) | Natural depression in Aravalli hills (Est. 1947) | Largest natural lake in Haryana (~3,000 acres); fed by rain runoff |
| **Badkhal Lake** | Faridabad | Man-made bundh constructed in 1947 | Built under CM Bansi Lal / Nehru for agricultural irrigation; name from 'Be-dakhal' |
| **Brahma Sarovar** | Thanesar (Kurukshetra) | Ancient sacred tank mentioned by Al-Biruni in 'Kitab-ul-Hind' | Largest man-made bathing tank in Asia (1800m x 1400m); Solar Eclipse dip |
| **Sannihit Sarovar** | Kurukshetra | Meeting point of 7 sacred Saraswatis | Believed to hold waters of all sacred teerthas on Amavasya; Lord Vishnu seat |
| **Karna Lake & Chakravarty** | Karnal (NH-44) | Artificial lake named after warrior Karna | Major roadside tourism hub of Haryana Tourism; Uchana complex |
| **Tilyar Lake** | Rohtak | Spread over 132 acres with Zoo | Major tourist complex and recreation hub of central Haryana |
| **Bhindawas & Khaparwas** | Jhajjar district | Freshwater wetland lake | Bhindawas is Haryana's largest wetland lake and designated Ramsar Site (2021) |
| **Blue Bird Lake** | Hisar | Artificial recreation lake & wetland | Famous migratory bird wetland near deer park and civil aerodrome |

METEOROLOGICAL PROFILE

# Climatic Zones, Temperature Extremes & Rainfall Isohyets

*Key Exam Takeaway: Climate is sub-tropical continental monsoon; Chhachhrauli is the 'Cherrapunji of Haryana' (>1100 mm).*

## TEMPERATURE & SEASONAL DYNAMICS

• **Classification:** Sub-tropical continental, semi-arid to sub-humid monsoon climate with extreme continental variations.

• **Temperature Extremes:** Searing summer peaks (>46°C - 48°C) in Hisar, Bhiwani, Sirsa, and Narnaul; freezing winter lows (<1°C - 2°C) with occasional ground frost (Pala/पाला) damaging potato and mustard crops.

• **Loo Winds (लू):** Hot, dry, dust-laden western winds blowing across southwestern plains from May to June.

• **Western Disturbances (Mawat / मावठ):** Winter precipitation in January-February brought by Mediterranean depressions; critical lifeline for Rabi wheat and gram.

• **HPSC Climate Question:** Southwestern fringe borders Thar Desert; northeastern fringe borders humid sub-Himalayas.

## RAINFALL DISTRIBUTION & ISOHYET MAP

• **Isohyet Gradient:** Average annual rainfall declines sharply from ~1,100 mm in the northeast to <300 mm in the southwest.

• **Highest Rainfall Belt:** Panchkula, Ambala, and Yamunanagar. Chhachhrauli tehsil (Yamunanagar) records the highest rainfall in the state (>1100 mm), earning the title 'The Cherrapunji of Haryana'.

• **Lowest Rainfall Belt:** Sirsa, Bhiwani, and Mahendragarh (average <300 mm; frequent drought episodes).

• **Monsoon Window:** South-West Monsoon (Arabian Sea and Bay of Bengal branches) active from early July to September, delivering ~80% of total annual rain.

• **Food Safety & Storage Implication:** High humidity (>80%) during July-August in eastern granary districts (Karnal, Kurukshetra) accelerates aflatoxin (Aspergillus flavus) proliferation in stored grains.

FORESTRY METRICS

# Forest Cover Assessment: India State of Forest Report (ISFR)

*Key Exam Takeaway: Haryana has ~3.63% forest cover (1,603 sq km); Panchkula leads with 43.87% while Palwal/Fatehabad trail.*

| Forest Assessment Metric | Area (sq km) | Percentage of State Area | High-Yield Screening Exam Details |
|---|---|---|---|
| **Very Dense Forest (VDF)** | 28.00 sq km | 0.06% | Confined to Kalesar National Park (Yamunanagar) and high Morni ridgelines |
| **Moderately Dense Forest (MDF)** | 451.00 sq km | 1.02% | Distributed in Panchkula, Yamunanagar, Ambala, and Aravalli scrub tracts |
| **Open Forest (OF)** | 1,124.00 sq km | 2.54% | Largest forest category; agro-forestry belts, canal bundhs, roadside strips |
| **Total Recorded Forest Cover** | 1,603.00 sq km | 3.63% | One of India's lowest forest percentages; National Forest Policy target is 33% |
| **Tree Cover (Outside Recorded Forest)** | 1,425.00 sq km | 3.22% | Extensive Poplar and Eucalyptus farm forestry in Yamunanagar/Karnal |
| **Total Green Cover (Forest + Tree)** | 3,028.00 sq km | 6.85% | Combined forest and agro-forestry canopy in Haryana |
| **District with Highest Forest Cover** | Panchkula: 390.70 sq km | 43.87% | Leading district in absolute area and proportional percentage |
| **District with Lowest Forest Cover** | Palwal (13.7 sq km) & Fatehabad | 0.71% - 0.73% | Intensively cultivated agricultural plains with minimal natural canopy |

PROTECTED AREAS

# National Parks of Haryana: Sultanpur (Ramsar) & Kalesar (Sal)

*Key Exam Takeaway: Haryana has exactly 2 National Parks: Sultanpur (Gurugram) and Kalesar (Yamunanagar).*

## 1. SULTANPUR NATIONAL PARK (GURUGRAM)

• **Geographical Location:** Farrukhnagar block, Gurugram district; spread over ~1.43 sq km (352 acres).

• **Historical Pioneer:** Declared a sanctuary in 1971 through efforts of famous British ornithologist Peter Scott and Dr. Salim Ali; upgraded to National Park in 1991.

• **Ramsar Site Recognition:** Designated as a Ramsar Wetland of International Importance in August 2021.

• **Avian Paradise:** Critical wintering ground for over 250 migratory bird species flying the Central Asian Flyway (Siberian crane, Greater flamingo, Bar-headed goose, Northern pintail).

• **Key Flora:** Acacia nilotica (Kikar), Ziziphus, Ber, and aquatic emergent marsh vegetation.

## 2. KALESAR NATIONAL PARK (YAMUNANAGAR)

• **Geographical Location:** Eastern foothills of Shivaliks, Yamunanagar district; spread over ~46.82 sq km (11,570 acres).

• **Upgraded to National Park:** December 2003 (surrounded by contiguous Kalesar Wildlife Sanctuary).

• **Interstate Forest Corridor:** Contiguous with Rajaji National Park (Uttarakhand) and Simbalbara National Park (Himachal Pradesh).

• **Dominant Flora:** Pure stands of mature Sal (Shorea robusta) forest, Semul, Khair, and Shisham; the only natural Sal forest in Haryana.

• **Faunal Diversity:** Wild Asian elephants (visiting from Rajaji), Leopards, Barking deer, Sambar, Goral, King Cobra, and Red Junglefowl.

• **Historical Monument:** Ancient British-era Kalesar Dak Bungalow (1901) and historical Shiva temple.

SANCTUARIES NETWORK

# Wildlife Sanctuaries of Haryana: 8 Sanctuaries & 2 Reserves

*Key Exam Takeaway: Abubshahar (Sirsa) is largest (11,530 ha); Chhilchhila (Kurukshetra) is smallest (28 ha); Bhindawas is Ramsar.*

| Sanctuary / Reserve | District Location | Area (Hectares) | Key Protected Fauna & Exam Points |
|---|---|---|---|
| **1. Abubshahar Sanctuary** | Sirsa (Mandi Dabwali border) | 11,530 ha | Largest sanctuary in Haryana; prime habitat for Blackbuck (Kala Hiran) & Nilgai |
| **2. Bhindawas Sanctuary** | Jhajjar district | 411.55 ha | Man-made freshwater wetland; Ramsar Site (2021); ~250 bird species; largest wetland lake |
| **3. Khaparwas Sanctuary** | Jhajjar (1.5 km from Bhindawas) | 82.70 ha | Wetland sanctuary recognized for waterfowl protection; notified 1991 |
| **4. Kalesar Wildlife Sanctuary** | Yamunanagar district | 5,435 ha | Adjoins Kalesar NP; dense Sal & mixed forest; Leopard, Wild Boar, Chital corridor |
| **5. Bir Shikargah Sanctuary** | Pinjore (Panchkula district) | 767.30 ha | Houses the famous Vulture Conservation Breeding Centre (VCBC); Sambar, Barking Deer |
| **6. Khol Hi-Raitan Sanctuary** | Panchkula (Shivalik belt) | 4,883 ha | Critical Shivalik eco-fragile corridor connecting Bir Shikargah and Morni Hills |
| **7. Nahar Wildlife Sanctuary** | Kosli tehsil (Rewari district) | 211.35 ha | Famous for Blackbuck, Jackal, Fox, and Peafowl in dry scrub deciduous forest |
| **8. Chhilchhila Sanctuary** | Kurukshetra (near Pehowa) | 28.92 ha | Smallest wildlife sanctuary in Haryana; vital shallow lake nesting site for winter ducks |
| **Conservation Reserves** | Bir Bara Ban (Jind) & Saraswati (Kaithal/Kurukshetra) | Bir Bara Ban: 419 ha<br>Saraswati: 4,452 ha | Bir Bara Ban (Jind town green lung); Saraswati Reserve (Bir Sonty) 2nd largest forest patch |

Chief Wildlife Warden, Haryana; MoEFCC Gazette Notifications.

EX-SITU CONSERVATION

# Specialized Wildlife Breeding Centres: Location & Species Focus

*Key Exam Takeaway: Pinjore hosts Asia's first Jatayu Vulture Conservation Breeding Centre (VCBC).*

## AVIAN BREEDING HUBS (PANCHKULA)

• **Jatayu Vulture Conservation Breeding Centre (VCBC):** Pinjore, Panchkula (Est. 2001 in collaboration with BNHS). Pioneer in captive breeding of 3 critically endangered Gyps species: Oriental white-backed, Long-billed, and Slender-billed vultures.

• **Diclofenac Ban Catalyst:** Research at Pinjore proved veterinary non-steroidal anti-inflammatory drug (NSAID) Diclofenac caused catastrophic visceral gout and renal failure in vultures, leading to a nationwide ban.

• **Pheasant Breeding Centre:** Morni, Panchkula district (Est. 1992-93). Specializes in Red Junglefowl and Kalij pheasant conservation.

• **Red Junglefowl Genetic Purity:** Rigorous DNA testing to prevent hybridization with domestic poultry.

## FAUNAL BREEDING & REPTILE CENTRES

• **Chinkara Breeding Centre:** Kairu, Bhiwani district. Established for breeding Indian Gazelle (Gazella bennettii) adapted to dry sand dune habitats.

• **Peacock & Chinkara Breeding Centre:** Jhabua, Rewari district. Dedicated to Indian Peafowl (Pavo cristatus) and Chinkara protection in Aravalli scrub.

• **Crocodile Breeding Centre (Bhor Saidan):** Bhor Saidan village near Pehowa, Kurukshetra district (Est. 1981-82). Houses Mugger crocodile (Crocodylus palustris).

• **Blackbuck Breeding Centre:** Pipli Mini Zoo, Kurukshetra and Rohtak Zoo.

• **Mini Zoos of Haryana:** Pipli Zoo (Kurukshetra), Rohtak Zoo, and Bhiwani Zoo under CZA regulatory standards.

# Aravalli Green Wall Project, Urban Oxy-Vans & Climate Action

*Key Exam Takeaway: Aravalli Green Wall Project aims to afforest a 5 km buffer along the 1,400 km Aravalli belt.*

## ARAVALLI GREEN WALL & ECO-RESTORATION

• **Aravalli Green Wall Initiative:** Inaugurated at Tikli village in Gurugram on International Day of Forests (March 2023).

• **National Buffer Scale:** 1,400 km long and 5 km wide green belt covering Haryana, Rajasthan, Gujarat, and Delhi.

• **Core Objectives:** Arrest eastward sand ingress from Thar desert; rejuvenate degraded Aravalli water recharge zones; combat regional desertification under UNCCD commitments.

• **Species Selection:** Indigenous drought-resilient trees: Kikar, Dhak (Butea monosperma), Ronjh, Neem, Peepal, and wild Ber.

• **Aravalli Safari Park:** Planned 10,000-acre project in Gurugram and Nuh districts; slated as the world's largest curated safari outside Africa.

## OXY-VANS, PRAN VAYU PENSION & STUBBLE MANAGEMENT

• **Oxy-Van (ऑक्सी वन) Initiative:** Urban micro-forests conceptualized during post-COVID environmental push. Flagship 800-acre Oxy-Van established at Karnal along Western Yamuna Canal.

• **Panchkula Oxy-Van:** Bir Ghaggar forest patch developed with 7 dedicated thematic groves (Panchavati, Navagraha, Nakshatra, Sugandh Vatika).

• **Pran Vayu Devta Pension Scheme:** Unique state scheme providing annual lifelong pension (~Rs 2,750+) to caretakers of healthy trees older than 75 years.

• **Air Quality Action (CAQM):** Stringent Commission for Air Quality Management regulations in 14 NCR districts of Haryana regulating stubble burning (Parali / पराली) and industrial boilers.

• **Bio-Energy Policy:** Extensive setup of 2G Ethanol Plant by IOCL at Panipat utilizing paddy straw to tackle seasonal smog.

# Vedic Civilization: Brahmavarta, Saraswati Basin & Kurukshetra

*Key Exam Takeaway: Manu Smriti defines Brahmavarta as the land between sacred rivers Saraswati & Drishadvati.*

## BRAHMAVARTA & VEDIC GEOGRAPHY

• **Brahmavarta (ब्रह्मावर्त):** Defined in Manu Smriti as the holy tract created by the gods between the divine rivers Saraswati and Drishadvati (Chautang).

• **Rigvedic Composition:** The majority of Rigvedic hymns were composed by Vedic seers (Bharadwaja, Vashistha, Vishvamitra) on the banks of the sacred Saraswati river.

• **Brahmarshi Desha:** Tract including Kurukshetra, Matsya, Panchala, and Surasena regarded as the highest ethical standard.

• **Kurukshetra Naming:** Named after King Kuru of Bharata dynasty, who ploughed the field with a golden plough across 48 kos (48 Kos Parikrama) to cultivate eight moral virtues (Satya, Dharma, Shaucha, etc.).

• **Jyotisar (ज्योतिसर):** The sacred banyan tree site where Lord Krishna delivered the celestial message of the Bhagavad Gita to Arjuna before the 18-day Mahabharata War.

## EPIC MAHABHARATA LANDSCAPE

• **48 Kos Parikrama:** Circumambulation encompassing over 134 sacred tirthas spread across Kurukshetra, Kaithal, Karnal, and Jind districts.

• **Karnal Connection:** Named after Anga-raj Karna, famous for unshakeable charity; Karna Lake and Karna Kot.

• **Prithudaka (Pehowa):** Ancient pilgrimage town on Saraswati river named after King Prithu; premier center in northern India for Pitru Tarpan / ancestral shradh rites.

• **Ban Ganga (Narkatari):** Site near Kurukshetra where Arjuna shot an arrow into the earth to quench Bhishma Pitamah's thirst on the bed of arrows.

• **Abhimanyu Pur (Amin):** Ancient village where 16-year-old warrior Abhimanyu was trapped in the Chakravyuha; recently renamed Abhimanyupur.

Rigveda; Manu Smriti (Chapter 2, Verse 17); Archaeological Survey of India (ASI).

# Rakhigarhi (Hisar): Largest Harappan Metropolis in South Asia

*Key Exam Takeaway: Spanning ~550 hectares across 11 mounds, Rakhigarhi is larger than Mohenjo-daro.*

## EXCAVATION HISTORY & URBAN SCALE

- **Geographical Setting:** Located in Narnaund tehsil of Hisar district in the dry palaeochannel of the ancient Drishadvati (Chautang) river.
- **Discovery & Excavations:** First identified by Acharya Bhagwan Dev (1960s); extensively excavated by ASI under Dr. Amarendra Nath (1997-2000) and Prof. Vasant Shinde (Deccan College, 2011-2017).
- **Magnitude & Scale:** Comprises 11 distinct archaeological mounds covering ~550 hectares, officially surpassing Mohenjo-daro and Harappa as the largest Mature Harappan site.
- **Urban Masterplanning:** Mud-brick defensive perimeter walls, multi-roomed houses with planned bathrooms, baked brick soak pits, covered street drainage, and terracotta drainage pipes.
- **Mound 7 Cemetery:** Yielded over 60 burials (primary, secondary, and symbolic burials) with rich grave pottery and jasper/steatite beads.

## SCIENTIFIC DISCOVERIES & DNA BREAKTHROUGH

- **Landmark aDNA Discovery (Cell, 2019):** Ancient DNA extracted from female skeleton (I6113) at Mound 7 proved: Indigenous South Asian ancestry with zero Steppe pastoralist or Anatolian farmer genetic contribution.
- **Monumental Granary:** Massive multi-chambered mud-brick structure with lime and decomposed grass plaster, engineered for grain storage and preservation against moisture.
- **Jewellery Workshop:** Uncovered lapidary beads of carnelian, agate, jasper, gold foils, copper mirrors, and micro-steatite beads indicating major trade hub.
- **Gold Smelting & Metallurgy:** Furnaces with crucibles containing minute traces of refined gold and copper slag.
- **World-Class Site Museum:** Central ASI and State Government site museum with on-site hostel, interpretation center, and experiential gallery.

Archaeological Survey of India (Dr. Amarendra Nath & Prof. Vasant Shinde); Cell / Science DNA Studies (2019).

# Banawali, Bhirrana & Kunal: The Fatehabad Civilizational Cradle

*Key Exam Takeaway: Bhirrana is India's oldest Harappan site (~7500 BCE); Banawali yielded clay plough & barley.*

| Archaeological Site | District & River Basin | Chronological Antiquity | Key Excavated Artefacts & Screening Traps |
|---|---|---|---|
| **Bhirrana (भीड़ाना)** | Fatehabad (Ancient Saraswati bed) | Hakra Ware phase (c. 7500 BCE - 6000 BCE) | Oldest Harappan settlement in South Asia (C-14 dating); red ware pot shard featuring graffito of 'Dancing Girl' mimicking Mohenjo-daro bronze idol |
| **Banawali (बनावली)** | Fatehabad (14 km from Fatehabad town) | Pre-Harappan, Mature, and Post-Harappan | Excavated by Dr. R.S. Bisht (1974); terracotta toy plough (मिट्टी का हल); rich caches of barley grains (जौ); radial town planning layout without rigid grid |
| **Kunal (कुणाल)** | Fatehabad (Ratia tehsil on Saraswati) | Early/Pre-Harappan (c. 4000 BCE) | Excavated by J.S. Khatri & M. Acharya; two exquisite royal silver tiaras/crowns (चांदी के मुकुट); gold disc ornaments; bone tools; pit-dwellings preceding brick houses |
| **Agroha (अग्रोहा)** | Hisar (NH-9 highway) | Post-Harappan to Kushan/Gupta period | Capital of legendary King Agrasen (Agrawal community); coins bearing legend 'Agrodaka'; terracotta tablet with 7 musical notes (Sa Re Ga Ma Pa Dha Ni) |

ASI Excavation Reports (R.S. Bisht, L.S. Rao, J.S. Khatri & M. Acharya); Nature Scientific Reports.

# Mitathal, Siswal, Farmana (Daksh Khera) & Balu

*Key Exam Takeaway: Farmana revealed 70 Harappan burials; Balu yielded earliest evidence of garlic cultivation.*

## BHIWANI & HISAR PROTO-HISTORIC NETWORK

• **Mitathal (Bhiwani):** Excavated by Dr. Suraj Bhan (1968). Yielded Late Harappan copper hoard including copper axes, parashu, and rings; shows transition to Ochre Coloured Pottery (OCP).

• **Siswal (Hisar):** Excavated by Dr. Suraj Bhan (1970). Type-site for the distinct 'Siswal Culture' characterized by Kalibangan-I related painted pottery in non-alluvial dry zones.

• **Tigrana (Bhiwani):** Recent excavations revealing large-scale terracotta and semi-precious stone bead manufacturing workshop.

• **Naurangabad (Bhiwani):** Ancient Prakritnaket; coins, seals, and terracotta moulds of the Yaudheya Republic (योध्येय गणराज्य) bearing legend 'Yaudheya Ganasya Jaya'.

## FARMANA, BALU & AGRICULTURAL ANTIQUITY

• **Farmana / Daksh Khera (Rohtak):** Joint excavation by Deccan College (Prof. Vasant Shinde) and Research Institute for Humanity and Nature (Japan). Massive 70-grave cemetery, copper ornaments, and jasper stamp seals.

• **Balu (Kaithal):** Excavated by Kurukshetra University (Dr. U.V. Singh). Crucial archaeobotanical find: Earliest evidence of cultivated Garlic (Allium sativum), green gram, and mustard seeds.

• **Sugh / Shrughna (Yamunanagar):** Ancient fortified port city on Yamuna; punch-marked coins, terracotta figurines, and Ashokan Buddhist heritage.

• **Bhadra & Daulatpur (Kurukshetra):** Late Harappan sites yielding painted grey ware (PGW) and clay seals.

# The Topra Ashokan Pillar, Chaneti Stupa & Buddhist Heritage

*Key Exam Takeaway: Firoz Shah Tughlaq moved the Topra Ashokan Pillar (Yamunanagar) to Kotla, Delhi in 1356 CE.*

## TOPRA ASHOKAN PILLAR EDICTS

• **Topra Kalan (Yamunanagar):** Original site of Emperor Ashoka's monumental monolithic polished sandstone pillar (3rd century BCE) bearing 7 Major Pillar Edicts in Brahmi script.

• **Only Pillar with 7 Edicts:** Topra is the only Ashokan pillar containing the entire Seventh Pillar Edict summarizing Dhamma policy (other pillars contain only 6).

• **Transport to Delhi (1356 CE):** Sultan Firoz Shah Tughlaq transported this multi-ton pillar on a 42-wheeled carriage pulled by 8,400 men and floated it down the Yamuna on large boats to install it atop Firoz Shah Kotla, Delhi.

• **Topra Ashoka Edicts Park:** Newly established memorial park in Yamunanagar housing exact replicas of all 14 Ashokan Rock Edicts and Topra pillar replica.

## CHANETI STUPA & SHRUGHNA MONASTERIES

• **Chaneti Buddhist Stupa (Yamunanagar):** Located 3 km from Jagadhri; built in 3rd century BCE during Ashoka's reign using concentric baked bricks, mimicking Sanchi hemisphere style.

• **Sugh (Ancient Shrughna):** Major Buddhist monastery mentioned by Chinese pilgrim Xuanzang (Hiuen Tsang); recorded 5 monasteries with over 700 Buddhist monks in Shrughna.

• **Karnal Inscription:** Kharosthi script inscription discovered at Karnal dating to 1st century BCE / CE.

• **Adi Badri Stupa:** Excavated Buddhist monastery and stupa structures in Yamunanagar foothills highlighting cultural confluence.

• **HPSC History Trap:** Remember, Topra was originally in Ambala district; after 1989 reorganization, it falls squarely in Yamunanagar district.

Inscriptions of Asoka (Corpus Inscriptionum Indicarum); Shams-i Siraj Afif's 'Tarikh-i Firoz Shahi'.

# Pushyabhuti Dynasty: Harshavardhana, Thanesar & Xuanzang

*Key Exam Takeaway: Harsha ruled from Thanesar (Kurukshetra) before shifting his imperial capital to Kannauj.*

## PUSHYABHUTIS OF THANESAR

• **Dynasty Founder:** Pushyabhuti (worshipper of Lord Shiva) founded dynasty at Sthanishvara (modern Thanesar in Kurukshetra).

• **Prabhakarvardhana:** First powerful king; adopted title 'Paramabhattaraka Maharajadhiraja'; called 'Huna-harina-kesari' (a lion to the Huna deer).

• **Accession of Harsha (606 CE):** Ascended throne at age 16 following the murder of elder brother Rajyavardhana by Gauda King Shashanka.

• **Shifting Capital:** Unified crowns of Thanesar and Maukhari (Kannauj) and shifted imperial capital to Kannauj for administrative centrality.

• **Literary Sovereign:** Harsha himself authored three acclaimed Sanskrit plays: Ratnavali, Priyadarsika, and Nagananda.

## COURT CULTURE & ARCHAEOLOGICAL LEGACY

• **Banabhatta (Court Poet):** Authored 'Harshacharita' (first historical biography in Sanskrit) and 'Kadambari'; described the prosperity of Srikantha Janapada (Haryana).

• **Visit of Xuanzang (Hiuen Tsang):** Traveled across Haryana (634-644 CE); stayed in Thanesar; praised social harmony, agricultural fertility, and Buddhist viharas.

• **Defeat by Pulakeshin II:** Harsha's southward expansion was halted at the Narmada River by Chalukya king Pulakeshin II (recorded in Aihole Inscription by Ravikirti).

• **Harsha-ka-Tila (Thanesar):** Excavated multi-period cultural mound in Thanesar revealing Kushana, Gupta, Harsha, Rajput, and Sultanate architectural layers.

• **Sheikh Chilli's Tomb:** Adjacent Sufi monument in Thanesar often called 'Taj Mahal of Haryana'.

Banabhatta's 'Harshacharita'; Xuanzang's 'Si-Yu-Ki'; Archaeological Survey of India.

# Tomar Dynasty, Prithviraj Chauhan & The Battles of Tarain (1191-92)

*Key Exam Takeaway: First Battle of Tarain (1191): Prithviraj won; Second Battle (1192): Ghori won, founding Delhi Sultanate.*

| Historical Event / Dynasty | Year / Period | Protagonists / Builders | Key Screening Points & Monuments |
|---|---|---|---|
| **Tomar Dynasty Rule** | 8th - 12th Century CE | Anangpal Tomar I & II | Founded 'Dhillika' (Delhi); built Surajkund Sun Temple & Dam in Faridabad; Pehowa Inscription of Bhoja |
| **Asigarh Fort (Hansi)** | 12th Century CE | Prithviraj Chauhan III | Also called 'Sword Fort' (Asi = Sword); strategic military stronghold of Chauhan kingdom |
| **First Battle of Tarain** | 1191 CE (Tarori, Karnal) | Prithviraj Chauhan vs Muhammad Ghori | Prithviraj's Rajput confederacy routed Ghori; Govind Rai wounded Ghori; Ghori fled to Ghazni |
| **Second Battle of Tarain** | 1192 CE (Tarori, Karnal) | Muhammad Ghori vs Prithviraj Chauhan | Ghori used 10,000 mounted archers; Rajput forces defeated; Prithviraj captured near Saraswati river; paved way for Muslim rule |
| **Qutb-ud-din Aibak Campaign** | 1192 - 1206 CE | First Mamluk Sultan | Crushed local resistance in Hansi, Kohram, and Rewari; established Turkish military outposts |

SULTANATE ERA

# Delhi Sultanate: Razia Sultan at Kaithal, Balban & Firoz Shah

*Key Exam Takeaway: Razia Sultan was assassinated near Kaithal (1240 CE); Firoz Shah Tughlaq founded Hisar & Fatehabad.*

## RAZIA SULTAN, BALBAN & TIMUR'S SCOURGE

• **Razia Sultan's Assassination (14 Oct 1240):** Razia (first female Muslim monarch of Delhi) and husband Malik Altunia were captured and murdered by rebel forces near Kaithal; historical Tomb of Razia Sultan exists in Kaithal town.

• **Ghiyas-ud-din Balban:** Subdued ferocious Mewati guerrilla rebels in southern Haryana; deforested Gurgaon/Mewat forests around Delhi; built a cordon of frontier forts at Hansi, Rewari, and Narnaul.

• **Invasion of Timur (1398 CE):** Plundered Sirsa, Fatehabad, Aharwan, Tohana, and Asandh en route to Delhi; faced fierce resistance from Jat and Ahir peasants.

• **Yaudheya & Khanzada Chiefs:** Semi-autonomous regional chieftains maintaining de facto control over south Haryana countryside.

## FIROZ SHAH TUGHLAQ'S URBAN & CANAL ARCHITECTURE

• **Foundation of Hisar-e-Firoza (1354 CE):** Built by Sultan Firoz Shah Tughlaq as a fortified royal township; features Gujri Mahal (built for his beloved milkmaid Gujri), Lat ki Masjid, and Diwan-e-Aam.

• **Lat (Pillar) of Firoz Shah:** Ashokan-style sandstone monolithic pillar section erected in the courtyard of Hisar Fort complex.

• **Foundation of Fatehabad:** Founded in 1351 CE and named after his newborn son Fateh Khan; houses Humayun Mosque and an Ashokan pillar fragment.

• **Firozpur Jhirka:** Established in southern Mewat hills as an administrative post.

• **Agricultural Canal Legacy:** Excavated original Western Yamuna Canal to supply water from Yamuna river to royal gardens in Hisar.

# The Three Battles of Panipat (1526, 1556, 1761): Strategic Analysis

*Key Exam Takeaway: Panipat altered Indian history thrice: 1526 (Mughal advent), 1556 (Mughal restoration), 1761 (Maratha decimation).*

| Parameter | First Battle of Panipat | Second Battle of Panipat | Third Battle of Panipat |
|---|---|---|---|
| **Date of Conflict** | 21 April 1526 | 5 November 1556 | 14 January 1761 (Makar Sankranti) |
| **Opposing Forces** | Zahir-ud-din Babur vs Sultan Ibrahim Lodi | Akbar (Bairam Khan) vs Hemu (Vikramaditya) | Ahmad Shah Abdali vs Sadashivrao Bhau (Marathas) |
| **Military Innovation** | Tulughma flanking tactic & Rumi cart-gun carriage (Araba) | Artillery bombardment; stray arrow hit Hemu's eye while atop 'Hawai' elephant | Abdali's mobile swivel camel-mounted guns (Zamburak) & cavalry |
| **Casualties / Decisive Event** | Ibrahim Lodi killed on battlefield (Tomb in Panipat) | Hemu captured unconscious & beheaded by Bairam Khan | 28,000+ Marathas slain; Vishwasrao & Jankoji Scindia martyred |
| **Historical Outcome** | Fall of Delhi Sultanate; foundation of Mughal Empire | Decisive consolidation of Mughal rule under Akbar | Shattered Maratha dream of Pan-Indian rule; paved way for British |
| **Memorial Landmark** | Kabuli Bagh Mosque (built by Babur in Panipat) | Hemu's Samadhi Sthal at Shodapur village | Kala Amb (Black Mango Tree) Memorial in Panipat |

Baburnama; Abul Fazl's 'Akbarnama'; J.N. Sarkar's 'Fall of the Mughal Empire'; Panipat Museum.

LAST HINDU EMPEROR

# Hemu (Samrat Hemchandra Vikramaditya): Rise from Rewari to Delhi

*Key Exam Takeaway: Hemu won 22 consecutive battles and was crowned Emperor at Purana Qila, Delhi (Oct 7, 1556).*

## ASCENT OF A NATIVE HERO

- **Humble Beginnings:** Born in a Brahmin/Dhussar merchant family at Machhari village (Alwar) and brought up in Qutabpur, Rewari.
- **Mercantile to Military Genius:** Started as a saltpetre trader (vital ingredient for gunpowder); recognized for brilliant logistics by Sultan Islam Shah Suri.
- **Prime Minister & Commander-in-Chief:** Appointed Wazir and supreme military general by Sultan Adil Shah Suri; won 22 consecutive battles against Afghan rebels and Mughal commanders.
- **Capture of Delhi & Agra (1556):** Smashed Mughal Governor Tardi Beg Khan at the Battle of Tughlaqabad (Delhi) on 6 October 1556.
- **Imperial Coronation:** Assumed imperial title 'Samrat Hemchandra Vikramaditya' at Purana Qila, Delhi on 7 October 1556; re-established indigenous Hindu sovereignty.

## BATTLE OF PANIPAT II & HISTORICAL LEGACY

- **The Tragic Turn at Panipat:** On 5 November 1556, Hemu's forces were routing the Mughal army under Bairam Khan when a rogue stray arrow pierced Hemu's right eye, causing him to fall unconscious in his howdah.
- **Mughal Vengeance:** Captured unconscious and brought before the 13-year-old Akbar; beheaded by Bairam Khan; Hemu's head was sent to Kabul and torso hung outside Delhi gate.
- **Father's Martyrdom:** Hemu's 80-year-old father Puran Das refused to convert and was martyred at Alwar.
- **Historical Stature:** Described by historians as the 'Napoleon of Medieval India' and the last Hindu sovereign to hold Delhi's imperial throne.
- **Memorial:** Hemu's Samadhi Sthal situated at Shodapur village near Panipat.

Vincent Smith, 'Akbar the Great Mogul'; K.K. Bhardwaj, 'Hemu: Napoleon of Medieval India'.

# Battle of Karnal (1739) & The Adventurer Raja: George Thomas at Hansi

*Key Exam Takeaway: Nadir Shah defeated Muhammad Shah at Karnal; Irish adventurer George Thomas ruled Hansi (1797-1801).*

## BATTLE OF KARNAL (1739) & COLLAPSE

- **Battle Date:** 24 February 1739 at Karnal plains.
- **Opponents:** Nadir Shah (Shah of Persia / Afsharid dynasty) vs Mughal Emperor Muhammad Shah 'Rangeela'.
- **Military Disaster:** In just 3 hours of combat, Nadir Shah's disciplined musketeers annihilated the superior-numbered Mughal army; Burhan-ul-Mulk captured.
- **Plunder of Delhi:** Nadir Shah marched to Delhi, ordered horrific general massacre (Qatl-e-Aam), and looted the Peacock Throne (Takht-e-Taus) and the Koh-i-Noor diamond.
- **Treaty of Surji-Anjangaon (30 Dec 1803):** Following Second Anglo-Maratha War, Daulat Rao Scindia ceded Haryana territory to British East India Company.

## GEORGE THOMAS: THE JAHAZI SAHIB OF HANSI

- **George Thomas ('Jahazi Sahib'):** Irish deserter who rose through mercenary ranks under Begum Samru of Sardhana and Maratha commander Appa Khande Rao.
- **Independent Kingdom at Hansi (1797-1801):** Established independent rule carving out 800 villages; rebuilt Asigarh Fort at Hansi as his royal capital.
- **Mint & Artillery Foundry:** Established his own currency mint issuing 'Sikka-e-Sahib' coins and cast European-style cannons in Hansi.
- **Downfall:** Besieged by French General Pierre Perron commanding Maratha forces; surrendered Hansi in 1802; died en route to Calcutta at Berhampore.
- **Hansi Heritage:** George Thomas's residence 'Jahaz Kothi' (so called because it resembled a ship) preserved as heritage monument in Hisar.

# The 1857 Uprising: Outbreak at Ambala Cantonment (10 May 1857)

*Key Exam Takeaway: The uprising broke out at Ambala Cantt 9 hours BEFORE Meerut (8-9 hours earlier on 10 May 1857).*

## AMBALA CANTT: THE FORGOTTEN EPICENTER

- **Pioneering Revolt Timing:** Mutiny broke out in the 5th and 60th Native Infantry at Ambala Cantonment at ~9:00 AM on Sunday, 10 May 1857 (almost 9 hours prior to the Meerut rebellion at 5:00 PM).
- **The Traitor Sham Singh:** Sepoy Sham Singh betrayed the mutineers by leaking plan details to British officers, allowing commander General Anson to disarm the regiments.
- **Ambala War Memorial:** Grand 22-acre international-standard memorial museum constructed on NH-44 at Ambala Cantt to honor unsung 1857 martyrs.
- **Strategic Ambala Cantt:** Established in 1843 (transferred from Karnal after malaria epidemic); northern military headquarters of British India.

## DISTRICT-WIDE SPREAD OF RESISTANCE

- **Regional Flare-Ups across Haryana:**
- - **Rohtak & Jhajjar:** Rebel sepoys led by Risaldar Bisarat Ali routed British district officers.
- - **Hisar & Hansi:** Led by Muhammad Azim (Prince of Delhi) and Hukam Chand Jain; British deputy commissioner John Wedderburn killed.
- - **Sirsa:** 4th Irregular Cavalry rebelled; British fled to Bikaner.
- - **Panipat:** Led by Imam Ali Qalandar and local Gujjar/Jat villagers cutting Grand Trunk Road communication.
- - **Mewat / Gurgaon:** Sadruddin Mewati led thousands of Meos against British forces; burnt police outposts.

State Archives Department, Haryana; Prof. K.C. Yadav, 'The Revolt of 1857 in Haryana'.

# Key Leaders of 1857: Rao Tula Ram, Raja Nahar Singh & Abdul Rahman

*Key Exam Takeaway: Rao Tula Ram liberated Rewari; Raja Nahar Singh ('Lion of Ballabhgarh') was hanged in Chandni Chowk.*

| 1857 Rebel Leader | Region / Princely State | Resistance Milestones | Fate & Commemoration |
|---|---|---|---|
| **Rao Tula Ram (with cousin Gopal Dev)** | Rewari & Ahirwal region | Liberated Rewari; established cannon factory at Rampura Fort; funded Mughal treasury at Delhi | Escaped to Iran/Afghanistan to seek Russian aid; died in Kabul (23 Sep 1863); Haryana Shaheedi Diwas |
| **Raja Nahar Singh ('Ballabhgarh Sher')** | Ballabhgarh (Faridabad) | Guarded Delhi's southern frontier; defied British commands; protected refugees | Arrested deceitfully; publicly hanged at Chandni Chowk on 9 Jan 1858; Ballabhgarh metro named in honor |
| **Nawab Abdul Rahman Khan** | Jhajjar princely state | Mobilized cavalry and infantry in support of Emperor Bahadur Shah Zafar | Tried by military commission; hanged outside Red Fort on 23 Dec 1857; Jhajjar territory annexed |
| **Nawab Ahmad Ali** | Farrukhnagar (Gurugram) | Co-ordinated Meo and Ahir peasant resistance against British relief columns | Hanged at Kotwali Chandni Chowk on 23 Jan 1858 |
| **Lala Hukam Chand Jain & Muneer Beg** | Hansi (Hisar district) | Drafted plans to overthrow British garrisons across western Haryana | Hanged simultaneously from the same tree in Hansi in 1858 |

National Archives of India; Punjab Mutiny Records (1857-58); Haryana Sahitya Akademi.

# Battle of Naseepur (16 Nov 1857) & Haryana Shaheedi Diwas (23 Sep)

*Key Exam Takeaway: Haryana observes Shaheedi Diwas on 23 September in sacred memory of Rao Tula Ram.*

## THE BLOODBATH AT NASEEPUR

• **Battlefield Setting:** Naseepur plain situated 3 km north of Narnaul town (Mahendragarh district).

• **Date of Epic Battle:** 16 November 1857.

• **Allied Forces:** Joint patriotic command under Rao Tula Ram, Gopal Dev, Gen. Abdus Samad Khan (Jhajjar), and Jodhpur Legion mutineers (5,000+ fighters).

• **Opposing British Column:** Col. Gerrard leading combined force of 1st European Bengal Fusiliers, Sikh cavalry (Patiala/Jind), and Multani horse.

• **Battle Climax:** Fierce hand-to-hand combat; British Commander Col. Gerrard was shot dead on the field; Captain Morehead wounded; allied forces eventually succumbed to heavy artillery fire.

## SHAHEEDI DIWAS & TERRITORIAL REPERCUSSIONS

• **Haryana Shaheedi Diwas (23 September):** Observed as a gazetted State public holiday every year in Haryana to honor Rao Tula Ram who breathed his last in Kabul on 23 September 1863.

• **Post-1857 Disenfranchisement:** Under the Queen's Proclamation of 1858, Haryana was ripped away from North-Western Provinces and merged with Punjab as collective punishment.

• **Princely Rewards for Loyalist Chiefs:** Princely states that aided the British (Patiala, Nabha, Jind, Kapurthala) were rewarded with confiscated territories of Jhajjar, Dadri, and Narnaul.

• **Exemplary Ahirwal Valour:** Legacy continued through the Battle of Rezang La (1962) where 13 Kumaon (Ahir troops) made supreme sacrifice under Major Shaitan Singh.

PEASANT CRUSADER

# Sir Chhotu Ram (Deenbandhu / Rahbar-i-Azam): Peasant Legislation

*Key Exam Takeaway: Sir Chhotu Ram passed revolutionary agrarian debt cancellation acts & conceptualized Bhakra Dam.*

## BIOGRAPHICAL & POLITICAL LANDMARKS

• **Early Life & Identity:** Born Ram Richhpal on 24 November 1881 at Garhi Sampla (Rohtak district); earned law degree from Agra College.

• **Jat Gazette (1916):** Launched premier weekly newspaper 'Jat Gazette' from Rohtak in Urdu/Hindi to articulate rural peasant grievances.

• **Unionist Party (Zamindara League 1923):** Founded non-communal agrarian party with Sir Fazl-i-Hussain representing Hindu, Sikh, and Muslim agriculturists.

• **Bhakra Dam Visionary:** Spearheaded water agreement with the Raja of Bilaspur (1944) laying the legal and financial foundation for Bhakra Dam.

• **Iconic Statues:** Prime Minister Narendra Modi unveiled a monumental 64-foot statue of Sir Chhotu Ram at Garhi Sampla in 2018.

## LEGISLATIVE MAGNA CARTA FOR FARMERS

• **Landmark Pro-Farmer Legislative Acts:**

• **1. Punjab Relief of Indebtedness Act 1934:** Liquidated predatory moneylender debts; capped compound interest; protected oxen and agricultural tools from auction.

• **2. Punjab Debtor's Protection Act 1936:** Exempted ancestral agricultural land and standing crops from court attachment.

• **3. Punjab Restitution of Mortgaged Lands Act 1938:** Restored hundreds of thousands of mortgaged agricultural acres back to destitute farmers without cash payment.

• **4. Punjab Agricultural Produce Markets Act 1939:** Established institutional Mandis (Market Committees) eliminating fraudulent merchant weights (Dami).

• **Titles:** Conferred 'Rao Bahadur' (1919), Knighthood ('Sir' in 1937), 'Rahbar-i-Azam' by grateful peasants.

Dr. Raghuvendra Tanwar, 'Sir Chhotu Ram: A Biography'; Punjab Legislative Assembly Debates.

# Freedom Struggle Stalwarts: Lala Lajpat Rai & Pandit Neki Ram Sharma

*Key Exam Takeaway: Pt. Neki Ram Sharma founded 'Sandesh'; declared 'I want the entire land of India'.*

| Freedom Fighter / Pioneer | Native Region | Major Publications / Institutions | Iconic Contribution / Exam Quotation |
|---|---|---|---|
| **Lala Lajpat Rai ('Punjab Kesari')** | Hisar (Karma-bhoomi 1886-92) | Founded Hisar Arya Samaj (1886) & Hisar Congress (1887) | Practised law at Hisar Court; served on Hisar Municipal Committee; established national protest networks |
| **Pandit Neki Ram Sharma ('Haryana Kesari')** | Kelanga village (Rohtak/Bhiwani) | Weekly Newspaper: 'Sandesh' (संदेश) from Bhiwani | Opposed British recruitment in WWI; famously told Viceroy: 'Mujhe to sampoorna Bharat ki bhoomi chahiye' |
| **Lala Murli Dhar ('Grand Old Man of Punjab')** | Ambala (Born in Palwal) | Founding member of Indian National Congress (1885 Bombay) | Jailed in 1886; renounced title 'Rai Bahadur' & 'Kaisar-i-Hind' medal in protest against Jallianwala Bagh massacre (1919) |
| **Deshbandhu Gupta** | Panipat | Daily Tej (Delhi) & nationalist journalism | Championed integration of Haryana and Delhi in 1932; leading figure in Constituent Assembly |
| **Smt. Kasturba & Mahatma Gandhi Visits** | Palwal & Rohtak | First arrested in Haryana at Palwal Railway Station (10 Apr 1919) | En route to Punjab against Rowlatt Act; Gandhi Smarak established at Palwal railway platform |

# Haryana Legislative Assembly: Composition, Evolution & Reservation

*Key Exam Takeaway: Unicameral house of 90 seats; exactly 17 seats are reserved for Scheduled Castes (SC).*

## ASSEMBLY STRUCTURE & SEAT EVOLUTION

• **Unicameral Architecture:** Single legislative chamber (Haryana Vidhan Sabha) located in the Capitol Complex, Chandigarh (designed by Le Corbusier).

• **Evolution of Assembly Seats:**

• • **At State Formation (1 Nov 1966):** 54 seats (10 reserved for SC).

• • **First General Elections (March 1967):** Increased to 81 seats (15 reserved for SC).

• • **Delimitation of 1977:** Increased to 90 seats (17 reserved for SC); remains constant until 2026 freeze.

• **Proposed New Assembly Building:** 10 acres of land allocated near Railway Station light point in Chandigarh for expanded delimitation capacity (>126 seats).

## 17 RESERVED CONSTITUENCIES (SC) & EXCEPTIONS

• **17 Reserved Constituencies (SC):**

• 1. Kalka/Mulana (Ambala)  2. Sadhaura (Yamunanagar)

• 3. Shahbad (Kurukshetra)  4. Guhla (Kaithal)

• 5. Nilokheri (Karnal)  6. Israna (Panipat)

• 7. Kharkhoda (Sonipat)  8. Narwana (Jind)

• 9. Uklana (Hisar)  10. Bawanikhera (Bhiwani)

• 11. Kalanaur (Rohtak)  12. Jhajjar (Jhajjar)

• 13. Bawal (Rewari)  14. Pataudi (Gurugram)

• 15. Hodal (Palwal)  16. Rania/Kalanwali (Sirsa)

• 17. Ratia (Fatehabad)

• **Districts with ZERO Reserved Seats:** Panchkula, Charkhi Dadri, Mahendragarh, and Nuh (Mewat).

PARLIAMENTARY PROFILE

# Parliamentary Representation: 10 Lok Sabha & 5 Rajya Sabha Seats

*Key Exam Takeaway: Haryana sends 10 MPs to Lok Sabha (2 reserved: Ambala & Sirsa) and 5 MPs to Rajya Sabha.*

| Constituency Type | Total Seats | Specific Constituencies | Key Screening & Electoral Details |
|---|---|---|---|
| **Lok Sabha (Lower House)** | 10 Seats | 1. Ambala (SC) 2. Kurukshetra<br>3. Karnal 4. Sonipat<br>5. Rohtak 6. Sirsa (SC)<br>7. Hisar 8. Bhiwani-Mahendragarh<br>9. Gurgaon 10. Faridabad | 2 Reserved Seats for SC:<br>• Ambala (SC)<br>• Sirsa (SC)<br>Gurgaon is largest electorate; Sonipat & Karnal major agricultural hubs. |
| **Rajya Sabha (Upper House)** | 5 Seats | Elected by 90 MLAs of Haryana Vidhan Sabha through proportional representation (STV) | Staggered 6-year tenure; one-third members retire every two years; represents state interest at Center. |
| **Electoral Demographics** | 22 Districts | Over 2.01 Crore registered voters in Haryana electorate | Highest voter population in Badshahpur assembly (Gurugram); lowest in Narnaul. |
| **Electoral Boundaries** | Delimitation 2008 | Merged Mahendragarh with Bhiwani; carved out modern Gurgaon and Faridabad PC | Assembly segments per Lok Sabha constituency: Exactly 9 Assembly segments each (9 x 10 = 90). |

Election Commission of India (ECI); Lok Sabha Secretariat.

EXECUTIVE LEADERSHIP

# Chief Ministers of Haryana: Historic Tenures & Landmark Milestones

*Key Exam Takeaway: Pt. BD Sharma was 1st CM; Rao Birender Singh was 1st non-Congress CM; Bhajan Lal served longest.*

| Chief Minister | Party / Alignment | Tenure Milestone | Key Historical / Exam Fact |
|---|---|---|---|
| **Pt. Bhagwat Dayal Sharma** | Indian National Congress | 1 Nov 1966 - 23 Mar 1967 | First Chief Minister of Haryana; hailed from Beri (Jhajjar); later Governor of Odisha & MP. |
| **Rao Birender Singh** | Vishal Haryana Party (VHP) | 24 Mar 1967 - 2 Nov 1967 | First non-Congress CM; first elected male Speaker of Vidhan Sabha; founded regional Vishal Haryana Party (Symbol: Rising Sun). |
| **Bansi Lal (3 Non-Consecutive Eras)** | Congress / Haryana Vikas Party | 1968-75, 1985-87, 1996-99 (~4,268 total days) | 'Architect of Modern Haryana'; 100% rural electrification by Nov 1970; lift irrigation; alcohol prohibition (1996-98). |
| **Banarsi Das Gupta** | Congress | 1975-77 & 1990 | Served as Speaker, Deputy CM, and Chief Minister of Haryana. |
| **Chaudhary Devi Lal ('Tau')** | Janata Party / Lok Dal | 1977-79 & 1987-89 | 'King Maker'; twice Deputy Prime Minister of India; introduced Old Age Pension & farm loan waiver. |
| **Bhajan Lal ('Ph.D of Politics')** | Congress / Haryana Janhit Congress | 1979-86 & 1991-96 (Longest cumulative tenure: 3,952 days) | Longest-serving Chief Minister of Haryana; mastered coalition engineering and non-Jat consolidation. |
| **Om Prakash Chautala** | INLD | Sworn in 5 times (1989, 90, 90, 91, 1999-2005) | Highest number of oath ceremonies as CM (5 times); shortest single tenure (6 days in July 1990). |
| **Bhupinder Singh Hooda** | Congress | 5 Mar 2005 - 19 Oct 2014 (Continuous 9 yrs 7 mos) | Longest continuous single stretch as Chief Minister; IMT Manesar and education hubs. |

Haryana Vidhan Sabha Secretariat; General Administration Department (GAD).

# Governors of Haryana & Legacies of Devi Lal and Bansi Lal

*Key Exam Takeaway: Dharma Vira was 1st Governor; B.N. Chakravarty served longest; O.P. Verma served shortest (5 days).*

## GUBERNATORIAL PROFILE & PRESIDENT'S RULE

• **First Governor:** Shri Dharma Vira (1 Nov 1966 - 14 Sep 1967).

• **Longest-Serving Governor:** B.N. Chakravarty (15 Sep 1967 - 26 Mar 1976; 8+ years; died in office; B.N. Chakravarty Canal named after him).

• **Shortest-Serving Governor:** Justice O.P. Verma (2 July 2004 - 7 July 2004; barely 5 days).

• **First Female Speaker:** Smt. Shanno Devi (first female speaker of any state legislature in India!).

• **President's Rule in Haryana (3 Times):**

• 1. 21 Nov 1967 - 21 May 1968 (Rao Birender Singh regime fall)

• 2. 30 Apr 1977 - 21 Jun 1977 (Post-Emergency dissolution)

• 3. 6 Apr 1991 - 23 Jul 1991 (Om Prakash Chautala regime fall)

## DEVI LAL VS BANSI LAL: MODERN ARCHITECTS

• **Chaudhary Devi Lal ('Jan Nayak' / 'Tau'):**

• • Born in Teja Khera (Sirsa); organized historic 'Nyaya Yudh' in 1985-87.

• • Deputy Prime Minister of India (1989-1991) under V.P. Singh & Chandra Shekhar.

• • Introduced pioneering Old Age Pension (वृद्धावस्था पेंशन) of Rs 100/month in 1987 (now Rs 3,000/month).

• • Waived rural farmer and artisan loans up to Rs 10,000.

• **Chaudhary Bansi Lal ('Iron Man of Haryana'):**

• • Born in Golagarh (Bhiwani); modernized state infrastructure.

• • Spearheaded 100% rural electrification achieved on 29 Nov 1970 (felicitated by Indira Gandhi at Ucchani, Karnal).

• • Pioneered tourism naming after birds (Blue Jay, Skylark, Magpie, Parakeet).

# Punjab & Haryana High Court & Panchayati Raj Governance

*Key Exam Takeaway: High Court began in Lahore (1919); Haryana was 2nd state to mandate educational criteria for Panchayats.*

## PUNJAB & HARYANA HIGH COURT

- **Judicial Lineage:** Established at Lahore on 21 March 1919 under Letters Patent issued by King George V.
- **First Indian Chief Justice:** Sir Shadi Lal of Rewari (appointed Chief Justice of Lahore High Court in 1920).
- **Post-Partition Shift:** Shifted to Shimla (Peterhoff) in 1947; relocated to Le Corbusier-designed building in Chandigarh in 1955.
- **First Chief Justice post-1966:** Justice Mehar Singh was Chief Justice when Haryana was carved out on 1 Nov 1966.
- **Sanctioned Strength:** 85 judges (64 permanent, 21 additional); shared jurisdiction over Punjab, Haryana, and UT Chandigarh.

## PANCHAYATI RAJ & EDUCATIONAL CRITERIA MILESTONE

- **Haryana Panchayati Raj Act 1994:** Passed on 22 April 1994 enforcing 73rd Constitutional Amendment (Three-tier: Gram Panchayat, Panchayat Samiti, Zila Parishad).
- **Historic Amendment (2015):** Mandated minimum educational qualification to contest Panchayat polls (10th pass for General males, 8th pass for SC males & women, 5th pass for SC women).
- **Mandatory Sanitation & Defaulter Check:** Must have a functional toilet at home; no power bill arrears; no cooperative loan default.
- **Upheld by Supreme Court:** Validated in landmark case *Rajbala v. State of Haryana (2015)*.
- **Panchayat 2.0 Reforms:** Inter-District Council, Gram Darshan portal, and 50% reservation for women in Panchayati Raj institutions.

Punjab & Haryana High Court; Haryana Panchayati Raj Act, 1994; Supreme Court (Rajbala v. State of Haryana, 2015).

# Agrarian Profile: Cropping Seasons, Land Utilization & Production

*Key Exam Takeaway: Haryana contributes ~15% of wheat & ~10% of rice to the National Central Foodgrain Pool.*

## LAND UTILIZATION & GRAIN DOMINANCE

• **Net Sown Area:** ~3.52 million hectares (>80% of total geographical area), reflecting extraordinary land cultivation efficiency (national average ~43%).

• **Cropping Intensity:** Approximately 185% - 190%, driven by intensive tubewell and canal irrigation networks.

• **Total Foodgrain Production:** Surpasses 180-185 lakh metric tonnes annually.

• **National Contribution:** Despite occupying only 1.34% of India's landmass, Haryana is the 2nd largest contributor of wheat and 3rd largest of rice to the Central Pool (PDS / FCI).

• **Gross Irrigated Area:** ~89% of gross cropped area is under assured irrigation (Tubewells ~55%, Canals ~44%).

## THREE CROPPING SEASONS OF HARYANA

• **Rabi Season (Winter / Asadhi - असाढ़ी):**

• • **Sowing:** Oct - Dec; Harvesting: April - May.

• • **Major Crops:** Wheat (Gehun), Mustard (Sarson), Gram (Chana), Barley (Jau), Potato.

• **Kharif Season (Monsoon / Sawani - सावणी):**

• • **Sowing:** June - July; Harvesting: Sep - Nov.

• • **Major Crops:** Paddy (Dhan), Cotton (Narma/Kapas), Pearl Millet (Bajra), Sugarcane, Maize.

• **Zaid Season (Summer / Jayad):**

• • **Sowing:** March - June (between Rabi and Kharif).

• • **Major Crops:** Fodder (Jowar/Barseem), Watermelon, Muskmelon, Cucumber, Seasonal vegetables.

CROP GEOGRAPHY

# Crop Belts: Taraori Basmati Rice, Wheat, Mustard, Bajra & Cotton

*Key Exam Takeaway: Karnal is the 'Rice Bowl of Haryana'; Taraori Basmati accounts for over 60% of India's Basmati exports.*

| Crop Category | Leading Districts | Key Agro-Economic Feature | FSO Food Science / Quality Standard |
|---|---|---|---|
| **Basmati Rice (Dhan)** | Karnal, Kaithal, Kurukshetra, Panipat (Rice Bowl) | Taraori Basmati (HBC-19) world famous for aroma & elongation; accounts for >60% of national Basmati exports | Geographical Indication (GI) tagged; 2-acetyl-1-pyrroline (2-AP) compound confers distinctive aroma |
| **Wheat (Gehun)** | Sirsa, Jind, Hisar, Karnal, Fatehabad | Sirsa leads in absolute wheat output; high-yielding varieties WH-1105, HD-2967, HD-3086 dominant | Subject to moisture limit (<12%) and foreign matter inspection under FSSAI cereals regulation |
| **Rapeseed & Mustard (Rai/Sarson)** | Mahendragarh, Bhiwani, Rewari, Charkhi Dadri | Mahendragarh is the premier mustard producing hub; cold-pressed oil extraction in local expellers | Erucic acid regulation (<2.0% in canola types; FSSAI blending limits); pungency due to allyl isothiocyanate |
| **Pearl Millet (Bajra)** | Bhiwani, Mahendragarh, Rewari, Hisar | Staple dryland cereal; highly drought-resilient; Haryana covers Bajra under Bhavantar Bharpayee Yojana | Rich in iron (Fe) and dietary fiber; classified as Nutri-Cereal under International Year of Millets |
| **Cotton (Narma / Kapas)** | Sirsa, Fatehabad, Hisar, Jind ('White Gold') | Sirsa is the undisputed cotton capital of Haryana; Bt-Cotton dominates >95% acreage | Cottonseed oil is refined for edible consumption; rich in tocopherols (Vitamin E) |
| **Sugarcane (Ganna)** | Yamunanagar, Karnal, Ambala, Sonipat | Yamunanagar leads in production; Saraswati Sugar Mills (Yamunanagar) is one of Asia's largest mills | Sucrose recovery percentage (~9.5% - 10.5%); jaggery (Gur) manufacturing cluster at Shahbad |

Agricultural & Processed Food Products Export Development Authority (APEDA); HAFED.

LIVESTOCK EXCELLENCE

# Murrah Buffalo ('Black Gold') & Indigenous Hariana Cattle

*Key Exam Takeaway: Murrah buffalo is Haryana's 'Black Gold' (Kala Sona); Jind is known as the 'Home of Murrah'.*

## MURRAH BUFFALO ('BLACK GOLD')

• **Origin & Homeland:** Rohtak, Jind, Hisar, and Fatehabad districts; Jind is reverently dubbed 'The Capital / Home of Murrah'.

• **Physical Conformation:** Jet-black color, tightly curled spiral horns (like जलेबी), wedge-shaped body, short legs, thin neck, and deep capacious udder.

• **Exceptional Milk Yield:** 2,000 - 3,500 kg per lactation; champions exceed 25-30 liters/day.

• **High Milk Fat Content:** Average fat 7.0% - 8.5%; preferred for ghee, paneer, and mawa production.

• **Global Sire Export:** Murrah bulls exported globally (Brazil, Italy, Bulgaria, Philippines) to upgrade native water buffalo stocks.

• **Economic Title:** Reverently hailed as 'Black Gold' (काला सोना) and 'The Tractor of Rural Economy'.

## HARIANA CATTLE & GLF HISAR

• **Hariana Cattle Breed (Bos indicus):**

• • Native dual-purpose cattle breed originating in Rohtak, Hisar, Jind, and Gurugram.

• • Bullocks renowned nationwide for unmatched draught power, endurance, and swift road transport.

• • **Coat:** White or light grey; long narrow face, flat forehead with convex center.

• **Sahiwal & Tharparkar Propagation:**

• • Promoted under Rashtriya Gokul Mission; Sahiwal cattle breeding farm at Government Livestock Farm (GLF) Hisar.

• **Government Livestock Farm (GLF), Hisar:**

• • One of the largest livestock breeding farms in Asia; established in 1809 as a camel breeding stud by Major James Lumsdaine.

• **Gau Seva Aayog:** Statutory Haryana Gau Seva Aayog headquartered at Panchkula regulating ~600+ Gaushalas.

RESEARCH ECOSYSTEM

# Premier Research Institutes: NDRI, CIRB, LUVAS & NBAGR

*Key Exam Takeaway: NDRI shifted from Bangalore to Karnal in 1955; cloned world-famous buffalo 'Garima' & 'Pratham'.*

| Research Institute | Headquarters | Establishment Year | Pioneering Research Achievements (FSO Must-Know) |
|---|---|---|---|
| **ICAR - National Dairy Research Institute (NDRI)** | Karnal | Est. 1923 (Bangalore)<br>Shifted to Karnal: 1955 | Deemed University; developed Karan-Swiss and Karan-Fries cattle crossbreeds; cloned buffalo 'Pratham' (1990) & 'Garima' (2009) using Hand-Guided Cloning; developed rapid chemical test kits for milk adulteration (detergents, starch, urea) |
| **ICAR - Central Institute for Research on Buffaloes (CIRB)** | Hisar (Sirsa Road) | 1 February 1985 | Premier institute for buffalo genetics, embryo transfer technology (ETT), and cryopreservation of semen; cloned superior Murrah bull 'Hisar-Gaurav' and 'Assamese' buffalo |
| **Lala Lajpat Rai University of Vet & Animal Sciences (LUVAS)** | Hisar | 2010 (Carved from CCSHAU) | State veterinary university; veterinary pathology, microbiology, zoonotic diseases, and animal disease diagnostic laboratories |
| **ICAR - National Bureau of Animal Genetic Resources (NBAGR)** | Karnal | 1984 (Shifted to Karnal 1985) | Nodal national agency for registration and characterization of new indigenous livestock and poultry breeds in India |
| **Central Sheep Breeding Farm (CSBF)** | Hisar | 1969-70 | Premier central farm breeding high-quality Corriedale and Sonadi rams to boost mutton and fine wool production |

ICAR-NDRI Karnal; ICAR-CIRB Hisar; LUVAS Hisar; ICAR-NBAGR Karnal.

# Food Science Hubs: NIFTEM Kundli, Mega Food Parks & HAFED

*Key Exam Takeaway: NIFTEM Kundli (Sonipat) is an Institute of National Importance under Ministry of Food Processing Industries.*

## NIFTEM KUNDLI: ACADEMIC EPICENTER

• **National Institute of Food Technology Entrepreneurship & Management (NIFTEM):**

• • Located at Kundli, Sonipat district (adjacent to Delhi border on NH-44).

• • Established by Ministry of Food Processing Industries (MoFPI); declared an 'Institute of National Importance' by Parliament under NIFTEM Act 2021.

• • **State-of-the-art Pilot Plants:** Dairy, Fruits & Vegetables, Meat & Poultry, Cereal & Extrusion, and Packaging Technology.

• • **Certified Food Testing Laboratories:** NABL-accredited and FSSAI-notified referral laboratory testing pesticide residues, mycotoxins, heavy metals, and adulterants.

• • **Incubation & Entrepreneurship:** Provides testing and pilot-scale scaleup facilities for food processing MSMEs across North India.

## MEGA FOOD PARKS & HAFED COOPERATIVE NETWORK

• **Mega Food Parks & Industrial Hubs:**

• • **Mega Food Park at Rai, Sonipat:** Houses large-scale fruit and vegetable dehydration, cold storage, and tetra-pack lines.

• • **Mega Food Park at Barhi, Sonipat:** Focuses on bakery, confectionery, and ready-to-eat (RTE) processing.

• • **Mega Food Park at Dabwali, Sirsa:** Promoted by private/cooperative sector for kinnow and citrus processing.

• • **HAFED Mega Food Park at IMT Rohtak:** Advanced milling, cold chains, and multi-crop automated packaging.

• **HAFED (Haryana State Co-op Supply & Marketing Fed):**

• • Largest apex cooperative body in Haryana (Est. 1 Nov 1966).

• • Operates automatic basmati rice mills at Taraori, Ding, and Rania; cold-pressed mustard oil mills at Rewari and Narnaul; cattle feed plants at Rohtak and Saktakhera.

Ministry of Food Processing Industries (MoFPI); HAFED Annual Report; NIFTEM Act, 2021.

HORTICULTURE INNOVATION

# Indo-Israel Centres of Excellence: Precision Farming & Processing

*Key Exam Takeaway: Gharaunda (Karnal) houses the Centre of Excellence for Vegetables; Mangiana (Sirsa) for Fruits.*

| Centre of Excellence (CoE) | District Location | Focal Commodity | Advanced Technology Implemented |
|---|---|---|---|
| **1. CoE for Vegetables** | Gharaunda (Karnal) | Tomato, Capsicum, Cucumber, Seedling nursery | Naturally ventilated polyhouses, climate-controlled hi-tech nurseries, soilless media (cocopeat), fertigation drip lines |
| **2. CoE for Fruits** | Mangiana (Sirsa district) | Kinnow, Pomegranate, Guava, Olive | Canopy management, high-density meadow orchards (guava), micro-irrigation, post-harvest fruit waxing and sorting |
| **3. CoE for Sub-Tropical Fruits** | Ladwa (Kurukshetra) | Mango, Pear, Litchi, Plum | Elite mother block preservation, rejuvenation of senile orchards, fruit fly trap management |
| **4. CoE for Integrated Beekeeping** | Ramnagar (Kurukshetra) | Honey & Bee Products (Apiculture) | Scientific queen bee rearing, honey processing and extraction facility, automated wax comb foundation plant |
| **5. CoE for Citrus Fruits** | Bhuna (Fatehabad district) | Sweet Orange (Malta), Lime, Kinnow | Viral-free certified citrus rootstock propagation, drip-fertigation monitoring |
| **6. CoE for Dairy / Protected Cultivation** | Hisar (CCSHAU campus) & Hodal (Palwal) | Dairy automation & Floriculture | Hydroponic automated fodder production, protected cultivation of gerbera and rose cut-flowers |

Mission for Integrated Development of Horticulture (MIDH); Indo-Israel Agricultural Project (IIAP).

# Industrial Powerhouses: Gurugram, Faridabad, Panipat & Yamunanagar

*Key Exam Takeaway: Gurugram produces ~50% of India's passenger cars; Panipat is the 'Textile / Weaver City'.*

| Industrial City / Hub | Key Economic Identity | Flagship Companies / Units | Key Manufacturing Output |
|---|---|---|---|
| **Gurugram (Millennium City)** | Automobile & IT Software Capital of North India | Maruti Suzuki (Manesar plant), Hero MotoCorp, DLF Cyber City, Google, Microsoft (>500 Fortune 500 offices) | Passenger cars, two-wheelers, financial technology, enterprise software, auto ancillary parts |
| **Faridabad (Industrial Hub)** | Oldest & largest heavy manufacturing city in Haryana | Escorts Group, JCB India, Yamaha Motors, Whirlpool, Goodyear Tyres, Havells | Tractors, earthmoving machinery, refrigerators, industrial rubber tyres, electrical switchgears |
| **Panipat (Textile / Weaver City)** | 'City of Weavers' & National Handloom Capital | IOCL Panipat Refinery (largest in South Asia), National Fertilizer Limited (NFL), Shodapur loom units | Military acrylic blankets (supplies 75% to armed forces), recycled yarn (shoddy wool), carpets, petrochemicals |
| **Yamunanagar (Paper / Timber City)** | National Timber Market & Paper Hub (Abdullapur Mandi) | Ballarpur Industries Limited (BILT), Saraswati Sugar Mills, Jagadhri Railway Workshop (Est. 1952) | Writing and printing paper, plywood (supplies 60% of India's plywood), railway wagons, refined sugar |

# Specialized Manufacturing Clusters: Rewari, Ambala, Bahadurgarh & Kundli

*Key Exam Takeaway: Rewari is the 'Brass City' (Pital Nagari); Ambala manufactures ~35% of India's scientific apparatus.*

## REWARI, AMBALA & BAHADURGARH

• **Rewari ('Brass City' / पीतल नगरी):** Century-old artisanal brass casting and sheet rolling; prominent manufacturer of brass and copper utensils, brass electrical components, and Tilla-jooti (embroidered footwear).

• **Jagadhri (Yamunanagar):** Renowned for stainless steel and brass household utensils; extensive rolling mills.

• **Ambala ('Scientific City' & 'Mixie City'):** Accounts for ~35% - 40% of India's total scientific laboratory equipment (microscopes, glassware, physics apparatus); also famous for domestic kitchen mixer-grinder manufacturing.

• **Bahadurgarh ('Gateway of Haryana', Jhajjar):** Major footwear hub in India (Relaxo, Action, Aero shoes); sanitaryware capital (Hindware / HSIL plants).

## IMT CORRIDORS: MANESAR, BAWAL & KHARKHODA

• **IMT Manesar (Gurugram):** Integrated Model Township spread over 4,500 acres; houses Maruti Suzuki, Denso, Honda Motorcycle, and National Brain Research Centre (NBRC).

• **IMT Bawal (Rewari):** Heavy automotive component and Japanese industrial corridor (JETRO) cluster on NH-48.

• **IMT Kharkhoda (Sonipat):** Massive 3,200-acre upcoming mega-manufacturing zone; new mega-plant of Maruti Suzuki (capacity 10 lakh cars/year) and Suzuki two-wheelers.

• **Kundli & Rai (Sonipat):** Food processing, stainless steel fabrication, and plastics manufacturing.

• **Dharuhera (Rewari):** Major automobile and industrial hub (Hero MotoCorp flagship assembly plant).

# Mineral Resources of Haryana: Mahendragarh Dominance & Distribution

*Key Exam Takeaway: Mahendragarh is the undisputed mineral capital of Haryana, holding deposits of iron, slate & calcite.*

| Mineral Commodity | Primary District / Geological Belt | Key Mining Localities | Industrial Application / Exam Fact |
|---|---|---|---|
| **Iron Ore (Magnetite/Hematite)** | Mahendragarh district | Antri-Beharipur, Dhanota-Dhancholi, Chhapra | Low to medium grade deposits associated with Aravalli quartzite ridges |
| **Slate Stone (स्लेट पत्थर)** | Rewari & Mahendragarh | Kund village (Rewari district) - World famous | Kund slate is exported globally for architectural roofing, paving, and chalkboards |
| **Limestone & Kankar** | Panchkula (Morni), Rohtak, Bhiwani | Malla, Kharag, and Tundapathar (Panchkula) | Feedstock for Surajpur Cement Factory (ACC - historical) and Charkhi Dadri cement plant |
| **Marble & Calcite** | Mahendragarh (Narnaul) | Bayal, Musnota, Mandi, Sirohi | Variegated marble (black, white, green banded) and pure crystalline calcite for filler industries |
| **Quartz & Silica Sand** | Gurugram, Faridabad, Mahendragarh | Aravalli ridges, Badarpur sand pits (Faridabad) | Glass manufacturing and construction concrete casting |
| **China Clay (Kaolin)** | Gurugram district | Sikanderpur and Alwar border | Sanitaryware, ceramic tiles, and porcelain insulators |
| **Flexible Sandstone (कम्पा पत्थर)** | Charkhi Dadri (Kaliana village) | Kaliana hillock | Unique geological curiosity: naturally bending/flexible stone (Itacolumite) found only here in India! |

DEMOGRAPHIC PROFILE

# Census 2011 Master Profile: Population, Density, Growth & Literacy

*Key Exam Takeaway: Total population: 2,53,51,462 (2.09% of India); Faridabad is most dense (2,442/sq km).*

| Census Demographic Parameter | Absolute Value / Figure | Highest District (Maximum) | Lowest District (Minimum) |
|---|---|---|---|
| **Total Population** | 2,53,51,462 (2.53 Crore)<br>(18th most populous in India) | Faridabad (18,09,733)<br>followed by Hisar | Panchkula (5,61,293)<br>followed by Charkhi Dadri (est) / Fatehabad |
| **Decadal Growth Rate (2001-2011)** | 19.90% (National: 17.70%) | Gurugram (+73.14%)<br>(Massive urban migration) | Jhajjar (+8.90%)<br>followed by Ambala (+11.23%) |
| **Population Density** | 573 persons / sq km<br>(National: 382/sq km) | Faridabad (2,442 persons/sq km)<br>followed by Gurugram (1,204) | Sirsa (303 persons/sq km)<br>(Large arid area, low settlement) |
| **Overall Literacy Rate** | 75.55% (National: 74.04%)<br>Male: 84.06% \| Female: 65.94% | Gurugram (84.70%)<br>followed by Panchkula (81.88%) | Nuh / Mewat (54.08%)<br>followed by Fatehabad (67.92%) |
| **Male Literacy** | 84.06% | Rewari (91.44%)<br>followed by Mahendragarh (89.72%) | Nuh / Mewat (69.94%)<br>followed by Fatehabad (76.14%) |
| **Female Literacy** | 65.94% | Gurugram (77.98%)<br>followed by Panchkula (75.99%) | Nuh / Mewat (36.60%)<br>(Severe socio-educational deficit) |
| **Urban-Rural Spread** | Rural: 65.12% (1.65 Cr)<br>Urban: 34.88% (88.42 Lakh) | Most Urbanized: Faridabad (79.51%)<br>followed by Gurugram (68.82%) | Least Urbanized: Nuh / Mewat (11.39%)<br>followed by Mahendragarh (14.41%) |

# Sex Ratio Dynamics, Child Sex Ratio & Beti Bachao Beti Padhao

*Key Exam Takeaway: Overall Sex Ratio in 2011 was 879 (lowest in India); PM launched 'BBBP' from Panipat in 2015.*

## CENSUS 2011 SEX RATIO REALITIES

• **2011 Overall Sex Ratio:** 879 females per 1,000 males (lowest among all 28 Indian states in Census 2011; national average was 943).

• **Highest Sex Ratio (2011):** Nuh / Mewat (907 females/1,000 males), followed by Fatehabad (902) and Rewari (898).

• **Lowest Sex Ratio (2011):** Gurugram (854 females/1,000 males) and Sonipat (856), reflecting severe urban selective abortions.

• **Child Sex Ratio (0-6 Years, 2011):** A catastrophic 834 females per 1,000 males (national average 918); Jhajjar (782) and Mahendragarh (775) were India's worst-affected districts.

• **Screening Question:** The district with highest overall sex ratio in 2011 is Nuh (907), while lowest is Gurugram (854).

## BETI BACHAO BETI PADHAO & RECOVERY

• **Historic Launch of 'Beti Bachao, Beti Padhao' (BBBP):**

• • Launched by Prime Minister Narendra Modi on 22 January 2015 from Panipat, Haryana.

• • Joint flagship mission of Ministry of Women & Child Development, Ministry of Health, and Ministry of Education.

• • **Strict Enforcement:** Severe crackdown on illegal sex determination under PC-PNDT Act (1994) with special task forces, covert sting operations, and reward schemes for informants.

• **Aapki Beti Hamari Beti Scheme:** Financial incentive deposited into LIC account on the birth of first girl child in SC/BPL families and second girl child in all families.

• **Civil Registration System (CRS) Resurgence:** Sex ratio at birth (SRB) improved significantly to ~915-920+ in recent years.

# Folk Theatre (Saang): Kishan Lal Bhat, Deep Chand & Dada Lakhmi Chand

*Key Exam Takeaway: Deep Chand Bahman is the 'Kalidasa / Shakespeare of Haryana'; Lakhmi Chand is 'Surya Kavi'.*

| Folk Theatre Master / Legend | Native Village & Period | Artistic Title / Honor | Pioneering Contribution & Famous Compositions |
|---|---|---|---|
| **Kishan Lal Bhat** | Meerut (Active c. 1730 CE) | Father / Originator of Saang format | Formalized open-air musical dramatic performance staged on wooden platform (Takhtaposh) with Sarangi and Nagada accompaniment |
| **Deep Chand Bahman** | Khanda village (Sonipat district)<br>Active WWI era | 'Kalidasa of Haryana'<br>'Shakespeare of Haryana' | Modernized Haryanvi Saang; introduced disciplined orchestra, musical meters (Alha, Chhand); recruited youth during WWI; authored 'Raja Harishchandra', 'Gopichand' |
| **Pandit Lakhmi Chand ('Dada Lakhmi')** | Jatti Kalan village (Sonipat)<br>(1903 - 1948) | 'Surya Kavi' (सूर्य कवि)<br>'Sufi Saint of Haryana' | Unlettered genius; composed over 25 classic Saangs (Nal-Damyanti, Satyavan-Savitri, Seth Taramal, Puranmal) and 500+ philosophical Ragnis; Research Chair in MDU Rohtak |
| **Pandit Mange Ram** | Sisana village (Sonipat district) | Chief Disciple of Lakhmi Chand | Authored acclaimed poetic Saangs: 'Pingla-Bhartari', 'Dushyant-Shakuntala'; mastered classical meter |
| **Baje Bhagat** | Sisana village (Sonipat district) | Renowned Ragni & Bhajan Master | Authored popular rural folk compositions: 'Raja Mordhwaj', 'Heer Ranjha'; moral aphorisms |
| **Ahmad Baksh Thanesari** | Thanesar (Kurukshetra) | Pioneering Epic Adaptor | Translated Valmiki Ramayana into Haryanvi Saang format; authored 'Gopi Chand' and 'Padmavat' |

# Folk Dances of Haryana: Dhamal, Phag, Khoria, Loor & Ratvai

*Key Exam Takeaway: Dhamal dates back to Mahabharata; Khoria is performed when groom departs; Loor means 'girl'.*

| Folk Dance Form | Cultural Occasion / Season | Gender / Performer Type | Distinctive Feature & Screening Trap |
|---|---|---|---|
| **1. Dhamal Dance (धमाल)** | Post-harvest moonlit nights; Spring | Exclusively Men (Ahirwal/Mahendragarh/Jhajjar) | Oldest dance form dating back to Mahabharata era; performed with Daph, Dholak, and wooden Chimta to celebrate good crop |
| **2. Phag Dance (फाग)** | Phalgun month (around Holi festival) | Men and women together (mixed) | Celebratory rhythm; witty banter with colored water sprinkling and rhythmic clapping |
| **3. Khoria Dance (खोरिया)** | Wedding night after Groom's Barat departs | Exclusively Women in domestic courtyard | Parody and joyful drama performed by women waiting for bride's arrival; men strictly prohibited |
| **4. Loor Dance (लूर)** | Holi season in Bangru linguistic belt | Exclusively Girls / Young Women | 'Loor' means girl in Bangru dialect; question-answer lyric format between two facing lines of dancers |
| **5. Chhathi Dance (छठी)** | 6th day after the birth of male child | Women of the household & neighborhood | Devotional thanksgiving to Bemata; boiled gram (Bakli / बाकली) distributed as prasad |
| **6. Gugga Dance (गुग्गा)** | Bhadrapada month (Gugga Navami) | Male devotees / Bhagats | Religious ritual dance; devotees self-flagellate with iron chains (Chhad/छड़) to honor snake deity Gugga Pir |
| **7. Ratvai Dance (रतवाई)** | Monsoon season in Mewat region | Meo men and women | Accompanied by Deru (small drum) and Bansuri; fast swirling circular formations |
| **8. Daph Dance (डफ)** | Spring harvest / Baisakhi celebration | Male dancers playing large circular frame drums | First state folk dance officially represented in Republic Day Parade (1969) under CM Bansi Lal |

# Traditional Attire, Folk Musical Instruments & Iconic Fairs

*Key Exam Takeaway: Surajkund Mela started in 1987; Khandwa is the turban of rural honor; Daaman is the 52-yard skirt.*

## TRADITIONAL ATTIRE & FOLK ORNAMENTS

• **Male Attire:** Khandwa (खंडवा - ceremonial turban symbolizing dignity and respect), Pagri / Safa, Kurta, and Dhoti.

• **Female Attire:** Daaman (दामण - pleated flared skirt traditionally made from 52 yards / बावन गज of heavy fabric), Kurti (fitted blouse), and Chundri (चूंदड़ी - printed veil).

• **Traditional Ornaments:**

• • **Head & Forehead:** Borla, Tika, Shringarpatti.

• • **Nose:** Nath, Koka, Purli.

• • **Neck:** Hansli (solid silver torque), Kathla, Galshree, Haar.

• • **Hands & Waist:** Pahunchi, Chhan-Kangan, Tagdi (silver waist chain).

• • **Feet:** Kari, Ramjhol, Pajeb.

## MUSICAL INSTRUMENTS & LANDMARK FAIRS

• **Folk Musical Instruments:** Ektara, Sarangi (associated with Saang), Been (snake charmer flute), Algoza (twin flutes), Deru, Nagada, Tasha, Khartal, and Chimta.

• **Surajkund International Crafts Mela (Faridabad):**

• • Annual fair held in February since 1987 in Surajkund Aravalli complex.

• • Features designated Theme State and Partner Country to promote rural crafts.

• **International Gita Mahotsav:** Held annually in Kurukshetra (Brahma Sarovar) on Shukla Ekadashi of Margashirsha (Mokshada Ekadashi).

• **Other Iconic Fairs:** Kapal Mochan Mela (Yamunanagar), Sheetla Mata Mela (Gurugram), Falgu Mela (Pharal, Kaithal - Somvati Amavasya), Mango Mela (Pinjore Gardens, July).

SPORTS POWERHOUSE

# Sports Capital of India: Olympic Medal Factory & Cash Award Policy

*Key Exam Takeaway: Haryana awards India's highest Olympic cash prizes (Gold: 6 Cr, Silver: 4 Cr, Bronze: 2.5 Cr).*

| Sports Discipline | Legendary Champions from Haryana | Native Village / District | Historic Milestone / Olympic Achievement |
|---|---|---|---|
| **Athletics (Javelin Throw)** | Subedar Major Neeraj Chopra (VSM) | Khandra village (Panipat district) | Independent India's 1st Track & Field Olympic Gold (Tokyo 2020: 87.58m); Olympic Silver (Paris 2024: 89.45m); World Champion |
| **Wrestling (Kushti)** | Sakshi Malik, Bajrang Punia, Phogat Sisters, Ravi Dahiya, Aman Sehrawat | Mokhra (Rohtak), Khudan (Jhajjar), Balali (Charkhi Dadri), Nahri (Sonipat) | Sakshi: 1st Indian female wrestler Olympic medal (Rio 2016 Bronze); Dahiya: Tokyo 2020 Silver; Aman: Paris 2024 Bronze |
| **Shooting** | Manu Bhaker | Goria village (Jhajjar district) | First Indian athlete post-independence to win 2 medals at a single Olympic Games (Paris 2024: 10m Air Pistol individual & mixed team) |
| **Boxing** | Vijender Singh, Amit Panghal, Nitu Ghanghas | Kaluwas (Bhiwani - 'Mini Cuba'), Mayna (Rohtak), Dhanana (Bhiwani) | Vijender: India's 1st Olympic Boxing medal (Beijing 2008 Bronze); Bhiwani Boxing Club (BBC) founded by Coach Jagdish Singh |
| **Field Hockey** | Rani Rampal, Savita Punia, Sandeep Singh, Surinder Kaur | Shahbad Markanda (Kurukshetra district) | Shahbad is renowned as the 'Nursery / Cradle of Indian Women's Hockey' & 'Sansarpur of Haryana' |

# District Profile Matrix Part I: Northern & Central Districts

*Key Exam Takeaway: Quick-revision encyclopedic matrix covering administrative hubs from Panchkula to Rohtak.*

| District | Creation Date | Historical / Nickname | Key Screening Points & Monuments |
|---|---|---|---|
| **Panchkula** | 15 Aug 1995 | 'City of Five Canals' (Panch Kula) | Morni Hills (Karoh 1514m); Pinjore Gardens (Yadavindra Gardens built by Fidai Khan); Mansa Devi Temple; HMT Pinjore; VCBC Vulture Centre |
| **Ambala** | 1 Nov 1966 (Original) | 'Scientific City' / 'Mixie City' | Ambala Cantt (Est. 1843); 1857 Uprising outbreak (May 10); birthplace of Sushma Swaraj; Gurdwara Manji Sahib; Muslim shrines |
| **Yamunanagar** | 1 Nov 1989 | 'Paper City' (Abdullapur Mandi) | Hathnikund & Tajewala; Kalesar NP; Saraswati Sugar Mills; Chaneti Stupa; Topra Ashokan pillar site; Jagadhri Railway Workshop |
| **Kurukshetra** | 23 Jan 1973 | 'Dharmakshetra' / Holy City | Brahma Sarovar & Sannihit Sarovar; Jyotisar (Gita sermon); Sheikh Chilli's Tomb; Kurukshetra University (1st Univ in Haryana, 1956); Kalpana Chawla Planetarium |
| **Kaithal** | 1 Nov 1989 | 'Kapishthala' (Hanuman birthplace) | Tomb of Razia Sultan (1240); Falgu Tirth (Pharal); Anjani Teela; Neem Sahib Gurdwara; Balu archaeological site |
| **Karnal** | 1 Nov 1966 (Original) | 'Rice Bowl' / 'City of Daanveer Karna' | NDRI (1955); CSSRI (1969); IIWBR; NBAGR; Karna Lake; Kalpana Chawla birthplace (1962); Liberty Shoes; Taraori Basmati export hub |
| **Panipat** | 1 Nov 1989 (Recreated 1992) | 'City of Weavers' / 'Textile City' | 3 Battles of Panipat (1526, 1556, 1761); Kabuli Bagh Mosque; Kala Amb; Bu Ali Shah Qalandar Dargah; Altaf Hussain Hali tomb; IOCL Refinery |
| **Sonipat** | 22 Dec 1972 | 'Swarnaprastha' / Golden City | NIFTEM Kundli; OP Jindal Global Univ; Sports University Rai (1st Sports Univ); Deenbandhu Chhotu Ram Univ (DCRUST Murthal); Atlas Cycles (historical) |
| **Rohtak** | 1 Nov 1966 (Original) | 'Rohitaka' / Education Hub | MDU Rohtak (Est. 1976); Pt. BD Sharma Post Graduate Institute of Medical Sciences (PGIMS); Tilyar Lake; Garhi Sampla (Chhotu Ram); Rewari-Rohtak eco zone |

DISTRICT ENCYCLOPEDIA II

# District Profile Matrix Part II: Western & Southern Districts

*Key Exam Takeaway: Quick-revision encyclopedic matrix covering administrative hubs from Hisar to Charkhi Dadri.*

| District | Creation Date | Historical / Nickname | Key Screening Points & Monuments |
|---|---|---|---|
| **Hisar** | 1 Nov 1966 (Original) | 'City of Steel' / Hisar-e-Firoza | Founded by Firoz Shah Tughlaq (1354); Gujri Mahal; CCSHAU (1970); CIRB (1985); LUVAS (2010); Jindal Steel; Rakhigarhi; Agroha Dham; Blue Bird Lake |
| **Fatehabad** | 15 Jul 1997 | 'Pink City of Haryana' | Founded by Firoz Shah (named after son Fateh Khan); Bhirrana (oldest site ~7500 BCE); Banawali & Kunal; Gorakhpur Nuclear Power Plant (2800 MW) |
| **Sirsa** | 1 Sep 1975 | 'Saraswati Nagari' / 'Arnyavanya' | Largest district by area (4,277 sq km); lowest population density (303); Abubshahar Sanctuary (largest); Ottu Barrage on Ghaggar; Tara Baba Kutiya |
| **Jind** | 1 Nov 1966 (Original) | 'Heart of Haryana' / Jayantipuri | Named after Jayanti Devi; 'Home of Murrah Buffalo'; Bir Bara Ban reserve; Dhamtan Sahib Gurdwara; Safidon (ancient Sarpadevi where Janamejaya performed snake sacrifice) |
| **Bhiwani** | 22 Dec 1972 | 'Chhoti Kashi' / 'Mini Cuba' | City of 300+ temples ('Chhoti Kashi'); Bhiwani Boxing Club; Mitathal & Naurangabad; Devsar & Star Monument; Board of School Education Haryana (BSEH) |
| **Charkhi Dadri** | 1 Dec 2016 | 22nd District of Haryana | Carved from Bhiwani; flexible sandstone (कम्पा पत्थर) at Kaliana; famous Phogat sisters (wrestling); 1996 mid-air plane crash site |
| **Mahendragarh** | 1 Nov 1966 (Original) | Mineral Capital of Haryana | Headquarters at Narnaul (only district whose HQ is not in its own name!); Dhosi Hill (652m, Chyavana Rishi); Birbal ka Chhatta; Jal Mahal; Antri-Beharipur iron |
| **Rewari** | 1 Nov 1989 | 'Brass City' / Ahirwal Epicenter | Rao Tula Ram (1857 hero); Hemu's hometown; Heritage Steam Locomotive Shed (1893, Fairy Queen); Kund slate stone; Bawal IMT; first Arya Samaj branch in Haryana (1880) |
| **Gurugram** | 1 Nov 1966 (Original) | 'Millennium City' / Guru Drona seat | Highest per capita income; Maruti Suzuki; Sheetla Mata Temple; Sultanpur NP; National Brain Research Centre (NBRC) Manesar; International Solar Alliance (ISA) HQ |
| **Faridabad** | 2 Aug 1979 | Industrial City (Baba Farid) | Most populous & dense district (2442/sq km); Surajkund crafts mela; Badkhal lake; Raja Nahar Singh Palace (Ballabhgarh); Escorts & JCB |
| **Palwal** | 15 Aug 2008 | 'Apelva' / Cotton City | Panchvati Temple; Mahatma Gandhi first arrested here (10 Apr 1919); Hodal Chameli forest; lowest recorded forest percentage (~0.71%) |
| **Nuh (Mewat)** | 4 Apr 2005 | Historical Mewat / Satyamevpuram | Highest sex ratio (907) & lowest literacy (54.08%); Kotla lake & Sheikh Musa tomb; Ratvai dance; Chhui Mal pond; Country's first Mobile Court (Punhana, 2007) |

# Landmark Initiatives: Parivar Pehchan Patra, MFMB & Bhavantar

*Key Exam Takeaway: Haryana is 1st state to implement Bhavantar Bharpayee Yojana (price deficit compensation).*

| Flagship Scheme / Portal | Launch Year | Core Objective | Screening Exam Question Focus |
|---|---|---|---|
| **Parivar Pehchan Patra (PPP)** | 2020 | 8-digit unique biometric/digital family ID for every Haryana household | Mandatory single identifier for availing all state DBT, pensions (Old Age), राशन कार्ड, and public scholarship schemes |
| **Meri Fasal Mera Byora (MFMB)** | 2019 | Unified digital portal for self-registration of cultivated crop acreage | Facilitates transparent MSP procurement by FCI/HAFED, crop loss compensation under PMFBY, and financial subsidies |
| **Bhavantar Bharpayee Yojana (BBY)** | 1 Jan 2018 | Price Deficit Financing Scheme for horticulture & agricultural crops | First state in India to launch BBY; compensates farmers if wholesale market price drops below designated benchmark price; covers 21+ horticulture crops & Bajra |
| **Mhara Gaon Jagmag Gaon (MGJG)** | 1 July 2015 | 24-hour uninterrupted three-phase electricity to rural Haryana villages | Launched from Dyalpura (Kurukshetra); linked to voluntary smart meter installation and power bill payment; >5,800+ villages covered |
| **Mukhyamantri Antyodaya Parivar Utthan** | 2021 | Targeted upliftment of poorest families earning <Rs 1.80 Lakh/year | Provides customized skill training, subsidized entrepreneurship loans, and self-employment linkages under Antyodaya Gramin Melas |
| **CM Window & Jan Samvad** | 25 Dec 2014 | Public grievance redressal system launched on Good Governance Day | Direct digital tracking of citizen complaints monitored at Chief Minister's Office (CMO) level |

FSO ENFORCEMENT DIRECTIVE

# Food Safety Administration in Haryana: Infrastructure & Enforcement

*Key Exam Takeaway: Food Safety Department functions under Food and Drugs Administration (FDA) Haryana, Panchkula.*

## STATUTORY ENFORCEMENT STRUCTURE

• **Apex Regulatory Body:** Food and Drugs Administration (FDA), Haryana (Headquartered at Sector-14, Panchkula).

• **Administrative Hierarchy:**

• • Commissioner of Food Safety (State Secretary / DG Health level).

• • Designated Officers (DO) at District Level (Administrative jurisdiction over licensing and prosecution).

• • Food Safety Officers (FSO) at Field Level (Inspection, sampling, seizures, and compounding).

• **Food Testing Infrastructure:**

• • **State Food Testing Laboratory, Karnal:** Premier NABL-accredited facility testing adulteration, pesticides, and microbial pathogens.

• • State Drug & Food Testing Lab, Chandigarh / Panchkula.

• • **Referral Testing:** Central Food Laboratory (CFL) Kolkata and NIFTEM Kundli.

## MOBILE TESTING & EAT RIGHT INITIATIVES

• **Mobile Food Testing Labs ('Food Safety on Wheels'):**

• • Fleet of customized mobile analytical vans operating across rural mandis, school mid-day meals, and dairy hubs.

• • Conducts instant on-spot testing for milk fat/SNF, synthetic milk (detergent, urea), starch in paneer, and adulterated edible oils.

• **Eat Right India Campaigns in Haryana:**

• • Eat Right Mela and Walkathons organized at Kurukshetra, Panchkula, and Gurugram.

• • **Clean Street Food Hubs:** Certified hygienic street food markets established at Sector-14 Panchkula and Gurugram.

• **Fortification Mandate:**

• • Fortified Wheat Flour (Atta fortified with Iron, Folic Acid, B12) distributed through Public Distribution System (PDS) across all 22 districts to combat stunting and anemia.

Food Safety and Standards Act 2006; Food and Drugs Administration (FDA) Haryana Annual Action Plan.

HIGH-YIELD REVISION

# Top 30 High-Frequency Exam Traps: Rapid-Fire Screening Checklist

*Key Exam Takeaway: Must-memorize high-yield factual checkpoints most frequently tested in HPSC & HSSC exams.*

## RAPID-FIRE CHECKPOINTS: 1 TO 15

- **1. Cherrapunji of Haryana:** Chhachhrauli (Yamunanagar).
- **2. Highest Elevation Peak:** Karoh Peak in Morni Hills (1,514 metres).
- **3. Highest Aravalli Peak:** Dhosi Hill in Mahendragarh (652 metres).
- **4. Oldest Harappan Settlement:** Bhirrana (Fatehabad, ~7500 BCE).
- **5. Largest Harappan Settlement:** Rakhigarhi (Hisar, ~550 hectares).
- **6. Clay Plough Excavated at:** Banawali (Fatehabad, by R.S. Bisht).
- **7. Royal Silver Crowns Excavated at:** Kunal (Fatehabad).
- **8. Earliest Garlic Evidence:** Balu (Kaithal).
- **9. Topra Ashokan Pillar Moved by:** Sultan Firoz Shah Tughlaq in 1356 CE.
- **10. 1857 Uprising Started at:** Ambala Cantt (10 May 1857, 9 AM).
- **11. Shaheedi Diwas Observed on:** 23 September (death of Rao Tula Ram).
- **12. First Chief Minister:** Pt. Bhagwat Dayal Sharma (Beri, Jhajjar).
- **13. First Non-Congress CM:** Rao Birender Singh (Vishal Haryana Party).
- **14. First Female Assembly Speaker:** Shanno Devi (1966).
- **15. First Indian Chief Justice:** Sir Shadi Lal (Lahore High Court, 1920).

## RAPID-FIRE CHECKPOINTS: 16 TO 30

- **16. 100% Rural Electrification Achieved on:** 29 Nov 1970 (under Bansi Lal).
- **17. Old Age Pension Introduced by:** Chaudhary Devi Lal in 1987 (Rs 100).
- **18. District with Highest Forest %:** Panchkula (43.87%).
- **19. District with Lowest Forest %:** Palwal (~0.71%) and Fatehabad.
- **20. Largest Wildlife Sanctuary:** Abubshahar (Sirsa, 11,530 ha).
- **21. Smallest Wildlife Sanctuary:** Chhilchhila (Kurukshetra, 28 ha).
- **22. Two Ramsar Wetland Sites:** Sultanpur NP (Gurugram) & Bhindawas (Jhajjar).
- **23. Vulture Breeding Centre Located at:** Bir Shikargah, Pinjore (Panchkula).
- **24. Home / Capital of Murrah Buffalo:** Jind district.
- **25. National Institute of Food Tech (NIFTEM) at:** Kundli (Sonipat).
- **26. Centre of Excellence for Vegetables:** Gharaunda (Karnal).
- **27. Flexible Sandstone (कम्पा पत्थर) at:** Kaliana (Charkhi Dadri).
- **28. Slate Stone Quarry Hub:** Kund village (Rewari).
- **29. Only District with HQ Outside Name:** Mahendragarh (HQ at Narnaul).
- **30. Beti Bachao Beti Padhao Launched on:** 22 Jan 2015 from Panipat.